MIMO-SST: Multi-Input Multi-Output Spatial-Spectral Transformer for Hyperspectral and Multispectral Image Fusion

Jian Fang, Jingxiang Yang, *Member, IEEE*, Abdolraheem Khader, *Member, IEEE*, and Liang Xiao, *Senior Member, IEEE*

***Abstract*— The current advanced hyperspectral super-resolution methods utilize convolutional neural networks (CNNs) that are either deeper or wider. These networks are designed to acquire end-to-end mapping capability, facilitating the transformation from low-resolution hyperspectral images (LR-HSIs) and high-resolution multispectral images (HR-MSIs) to high-resolution HSIs (HR-HSIs). The existing methods lack the capability to capture details and structures in the image effectively, while multi-input multi-output methods can address this issue efficiently. Therefore, this article proposes a novel network architecture named multi-input multi-output spatial-spectral transformer (MIMO-SST). To apply the multi-input multi-output methods in HSI fusion, specifically integrating the spatial-spectral information of LR-HSI and HR-MSI, we introduce multihead feature map attention, multihead feature channel attention, and a multiscale convolutional gated feedforward network, constructing the proposed mixture spatial-spectral Transformer. Moreover, to enhance the expressive power of image edges and recover the sharpened structure details, this study incorporates a novel wavelet-based high-frequency loss into the ultimate comprehensive loss, with the objective of refining the reconstruction of high-frequency details. Experimental studies on three simulated datasets and one real-world dataset demonstrate that the proposed method in this study outperforms contemporary state-of-the-art methods in terms of performance. It is noteworthy that our method exhibits a 0.85-dB improvement in terms of the peak signal-to-noise ratio (PSNR) metric on the Columbia computer vision laboratory (CAVE) dataset compared to state-of-the-art methods. Our code is publicly available at https://github.com/Freelancefangjian/MIMO-SST.**

***Index Terms*— Hyperspectral image (HSI) fusion, multi-input multi-output, Transformer.**

I. Introduction

THE hyperspectral image (HSI) is a 3-D image taken at different spectral bands, covering hundreds of continuous narrow bands in the visible to infrared spectrum [1], [2]. HSIs are rich in spectral information and are widely used in semantic segmentation [3], [4], scene classification [5], [6], target detection [7], [8], target tracking [9], [10], and other fields. Obtaining high-resolution (HR) images directly in the spatial and spectral domains is extremely difficult due to hardware and budget constraints. Therefore, the balance between spatial and spectral resolutions has been one of the significant challenges for HSI in practice [11], [12], [13], [14], [15]. A natural solution is to capture a low-resolution (LR) HSI and an HR multispectral image (MSI) and fuse them into one image with high spatial and spectral resolution. This process is known as HSI and MSI fusion (HSI–MSI fusion) [16].

In recent years, many approaches have been proposed to solve the HSI–MSI fusion problem from different perspectives. They can be broadly classified into two categories, namely, model-based methods [17], [18], [19], [20], [21], [22], [23], [24], [25] and deep learning (DL)-based methods [26], [27], [28], [29], [30], [31], [32], [33]. Regarding model-based approaches, many researchers have added different prior constraints to linear observation models to turn the HSI–MSI fusion problem into an ill-posed inverse problem. However, these manually prespecified priors usually achieve limited performance. In addition, the optimal algorithm parameters of these methods need to be tuned for different datasets. In recent years, DL-based methods, particularly those employing Transformer [34], [35], [36] technology, have been utilized to address the fusion problem of HSI and MSI. Network structures built using Transformer outperform methods based on convolutional neural networks (CNNs) in extracting non-local information. However, the end-to-end HSI fusion model with Transformer primarily focuses on global relationships, resulting in suboptimal performance for tasks emphasizing local information, such as HSI fusion. In contrast, adopting a multi-input multi-output approach aids in supervised HSI generation across various scales, enabling a more comprehensive capture of details and structures within the image. Smaller scales contribute to capturing local fine features, while larger scales aid in capturing global structures. This approach has been successfully applied in the domain of natural image processing, as documented in [37] and [38]. However, this approach cannot be directly applied to the HSI fusion domain.

We present a novel network architecture named multi-input multi-output spatial-spectral transformer (MIMO-SST),

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The authors are with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: fangjian@njust.edu.cn; yang123jx@njust.edu.cn; abdolraheem@njust.edu.cn; xiaoliang@mail.njust.edu.cn).

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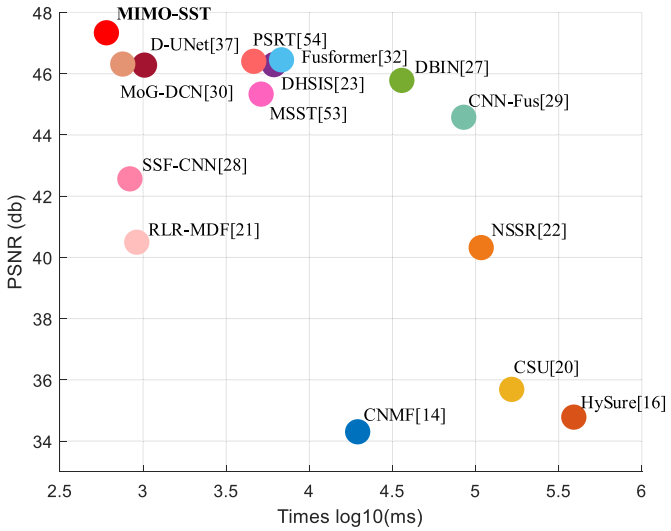


Fig. 1. Current investigation undertakes a comparative evaluation of the suggested techniques in relation to alternative methods, emphasizing two pivotal performance criteria, specifically, PSNR and runtime. The approach that we suggest is depicted prominently in the figure. The reported runtimes of these methods are obtained by measuring the execution times in our experiment environment using the published test code for each method.

designed to address the challenges of HSI–MSI fusion by leveraging the advantages of multiple inputs and outputs. To apply the multi-input multi-output methods to the field of HSI fusion, we introduce a mixture spectral-spatial Transformer to integrate the spectral information of LR-HSI and HR-MSI at different scales. The proposed mixture spectral-spatial Transformer incorporates multihead feature map attention (MFMA), multihead feature channel attention (MFCA), and multiscale convolutional gated feedforward networks (FNs). In addition, to enhance the network's perception of edges in the images and make the boundaries between objects clearer, we integrate a wavelet-based high-frequency (WHF) loss into the final total loss. Experimental results demonstrate that the proposed MIMO-SST method outperforms state-of-the-art methods in terms of peak signal-to-noise ratio (PSNR) and computational complexity, as illustrated in Fig. 1.

The main contributions of this article are listed in the following.

1) This research article introduces an advanced strategy for HSI–MSI fusion, employing multiple-input multiple-output (MIMO) processing techniques, specifically MIMO-SST. The proposed method utilizes LR-HIS and HR-MSI information at the detail level while leveraging information at its coarse and intermediate levels, addressing challenges in capturing details and structures in HSI. In addition, we use a multiscale loss tailored for the coarse-to-fine architecture to train the proposed model, greatly enhancing the convergence of the network.
2) We have created a mixture spatial-spectral Transformer (SST), constructed through a combination of MFMA, MFCA, and a multiscale convolutional gated FN. This approach exploits the correlation between LR-HSI and HR-MSI features using MFMA and MFCA. In addition, we introduce different scale convolutions to extract information from spatially adjacent pixel locations in the input features of the FN, aiding in the effective recovery of local image structures.
3) To enhance the expressive power of image edges and restore sharpened structural details, this study introduces a novel approach. This method incorporates a WHF loss into the overall loss function to improve the reconstruction of intricate high-frequency details.

The rest of this article is organized as follows. Section II offers a comprehensive account of the relevant literature. In Section III, we expound on the network methodology utilized in our study. Subsequently, we present the simulation experimental outcomes and analysis in Section IV, followed by the real data experimental findings and analysis presented in Section V. Finally, Section VI encapsulates the concluding remarks of our research.

II. Related Works

This section provides a review of some of the most notable recent research advances in HSI super-resolution techniques. Among the most relevant techniques are model-, DL-, multiscale DL-, and Transformer-based methods.

A. Model-Based

HSI super-resolution is a challenging problem characterized by its ill-posed nature. A common approach is to use priors or regularization to constrain the optimization equation. Several priors have been employed to regularize the HSI super-resolution problem [17], [19], [23]. Given the inherent redundancy in HSI, sparse priors can be utilized to constrain the spatial and spectral correlation of HSIs. Wei et al. [39] introduced a sparse regularization term based on a scene decomposition into a set of dictionaries. Lanaras et al. [23] proposed a generic Bayesian sparse coding strategy to be used with Bayesian dictionaries learned with the beta process. In addition, spectral unmixing priors are frequently used to address HSI super-resolution. In [40], a demixing-based formulation is employed and integrated with an explicit parametric model accounting for typical spectral variability between the two images. This combination aims to tackle the HSI super-resolution problem. Some methods utilize tensor-based approaches to address HSI super-resolution. In [41], a factor-smoothed regularization method for tensor ring (TR) decomposition is introduced to capture the spatial-spectral continuity of HR-HSI.

B. DL-Based

Early deep-learning fusion approaches were built on end-to-end networks that used different forms of convolutional deformation and network connectivity strategies to learn super-resolution mapping relationships from LR-HSI and HR-MSI to HR-HSI. The literature [27] uses residual learning, convolution, and multichannel convolution to jointly build networks to learn depth features to reconstruct HSI

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images. Model-driven DL methods are a new idea in current research. Such methods often combine forward processes and DL priors to improve the reconstruction performance of deep fusion networks and enhance the interpretability of the networks. Dian et al. [42] proposed a deep hyperspectral sharpening method in the literature, where they used deep convolutional residual networks to learn image prior information and build a minimization model for null-spectral fidelity based on degeneracy constraints to solve the problem. In response to the situation that supervised DL requires a large number of data samples to train network models, unsupervised DL fusion methods have received wide attention. For example, an unsupervised encoder–decoder architecture [43] has been proposed to solve the HSI super-resolution problem.

C. Multiscale DL-Based

Several researchers have proposed the fusion of LR-HSI and HR-MSI at different scales to obtain HR-HSI. Zhou et al. [44] proposed a pyramidal fully convolutional network (PFCN) to solve the HSI–MSI fusion problem. The network consists of two subnetworks: the first subnetwork extracts the spectral information of LR-HSI by convolution kernel and encodes it as depth features; the second subnetwork aims to combine the HR-MSI pyramid with the encoded depth features to obtain HR-HSI. The method proposed in [45] solves the fusion problem of HR-MSI and LR-HSI. In this network, the deep features of LR-HSI are gradually scaled up by deconvolution, and then, the deep features of LR-HSI and HR-MSI are fused at different scales. However, this structure ignores the basic and shallow features of MSI. Therefore, this work [46] introduces a dual-UNet fusion method, which first extracts MSI spatial features at different scales using an encoding–decoding network and then uses these scale features to inject them into the UNet. In order to address the limited flexibility of CNN with fixed square convolution kernels in handling irregular patterns and the issue of losing pixel-level features when using graph convolutional networks (GCNs) with superpixels to reduce the number of nodes, the attention multihop graph and multiscale convolutional fusion network (AMGCFN) is proposed in [47]. To address the limitation of existing studies, which only consider multiscale features containing changed and unchanged components, making it challenging to represent subtle changes between bitemporal HSIs at each scale, the proposed work in [48] introduces a multiscale diff-changed feature fusion network (MSDFFN) for HSI change detection. In their work [49], the authors introduced a deformable convolution alignment module that includes multiscale residual blocks. This module aims to address the alignment difficulties caused by limited motion and various scales of moving objects in remote-sensing images.

D. Transformer-Based

Transformer, a natural language processing (NLP) model for machine translation, was proposed by Vaswani et al. [50]. It has been introduced into computer vision in recent years and has gained popularity for its advantages in capturing long-distance correlations between spatial regions. Since directly replacing CNN convolutional layers with Transformer layers to process images would impose a huge computational burden, an early approach was to augment the standard CNN model with Transformer layers [51]. Recently, Chen et al. [52] proposed a Transformer-based backbone model for image recovery, image processing transformer (IPT), however, the number of parameters and the computational effort of IPT are very large, and it is complicated to train such a model. To address the problem of computational overload when using the Transformer for image processing tasks, it is proposed to segment the training image into a series of image patches [53]. However, the common Transformer cannot handle HR ones due to the LR feature maps and the fact that the complexity grows quadratically with size. A hierarchical Transformer represented by a shifted frame port calculation is proposed, also known as the Swin Transformer [54]. The Swin Transformer not only handles HR images like a CNN but also has the advantage of a Transformer to model long-range dependencies. In [55], an image recovery Transformer was developed based on the Swin Transformer, which achieves state-of-the-art performance in a variety of image reconstruction tasks. A custom Transformer architecture and information-neutral reversible neural modules for long-range dependency modeling and effective feature fusion are proposed in the literature [56]. In recent years, some scholars have applied the Transformer to the field of HSI processing, such as an improved SST [57] that combines migration learning and SST for HSI classification.

III. Methodology

In this section, we introduce the Transformer module, the spatial-spectral interactive Transformer structure, and the overall model structure. Let $\mathbf{X} \in \mathbb{R}^{H\times W\times b}$ denote the HR-MSI, where H, W, and b represent the dimensions of the spatial height, width, and spectral band number, respectively. Let $\mathbf{Y} \in \mathbb{R}^{h\times w\times B}$ denote the LR-HSI, where h, w, and B represent the dimensions of the spatial height, width, and spectral band number, respectively. As shown in Fig. 2, the MIMO-SST network structure is proposed in this article. This network structure consists of a mixture SST block (MTB), a multiple-input encoder, and a multiple-output decoder.

A. Mixture SST Block

Due to the complexity of this section, we divide it into four sections: feature extractor, MFMA, MFCA, and multiscale convolutional gated FN (MCGFN). In the following, we describe these sections in detail.

1) Feature Extractor: To emphasize contextual information, we introduce deep convolution before computing feature covariance and global attention. The spatial information and the spectral information of HR-MSI and upsampling LR-HSI are first extracted using convolution, and the extracted information is denoted by $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times C}$ and $\hat{\mathbf{Y}} \in \mathbb{R}^{H\times W\times C}$, respectively. Subsequently, we reshape $\hat{\mathbf{X}}$ and $\hat{\mathbf{Y}}$ to $\tilde{\mathbf{X}} \in \mathbb{R}^{HW\times C}$ and $\tilde{\mathbf{Y}} \in \mathbb{R}^{HW\times C}$, respectively. The three basic elements *query* $\hat{\mathbf{Q}} \in \mathbb{R}^{HW\times C}$, *key* $\hat{\mathbf{K}} \in \mathbb{R}^{HW\times C}$, and *value* $\hat{\mathbf{V}} \in \mathbb{R}^{HW\times C}$ of the MFMA and the three basic elements *query* $\widetilde{\mathbf{Q}} \in \mathbb{R}^{C\times HW}$, *key* $\widetilde{\mathbf{K}} \in \mathbb{R}^{C\times HW}$, and *value* $\widetilde{\mathbf{V}} \in \mathbb{R}^{C\times HW}$ of the

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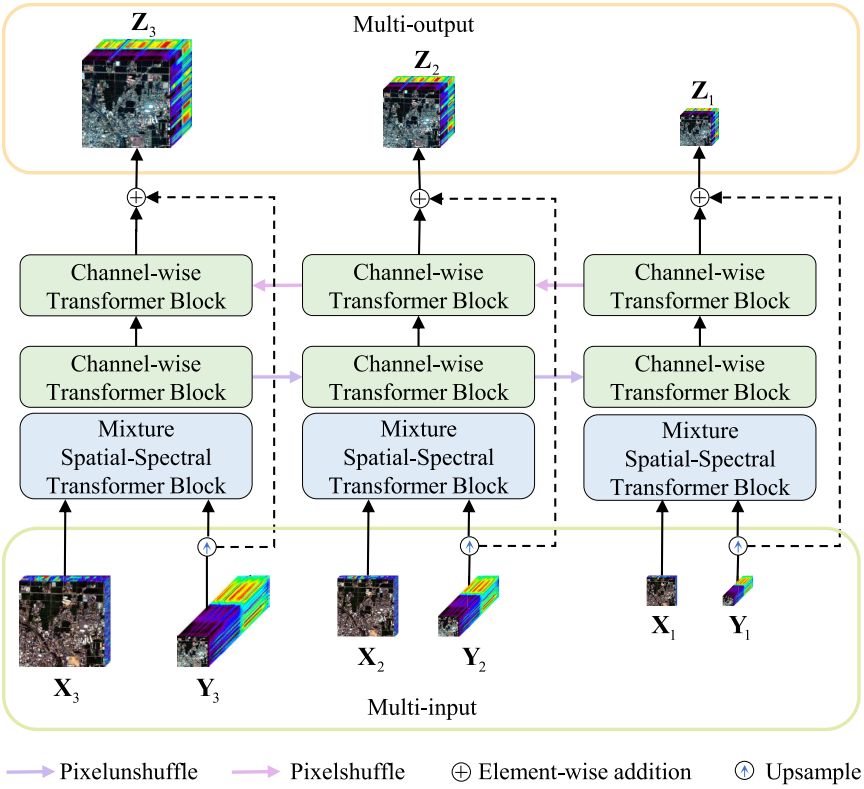


Fig. 2. MIMO-SST network proposed in this study is characterized by a comprehensive structure featuring six inputs and three outputs. Specifically, the network architecture comprises three HSIs (obtained through upsampling of the original HSI) and three MSIs (obtained via downsampling of the original MSI). Furthermore, the output results consist of three reference images corresponding to the image (which are obtained by means of downsampling), forming the basis for the calculation of the loss function.

MFCA are given as follows:

$$\hat{\mathbf{Q}} = W_d^q(W_p^q(\tilde{\mathbf{X}})) \tag{1}$$
$$\hat{\mathbf{K}} = W_d^k(W_p^k(\tilde{\mathbf{Y}})) \tag{2}$$
$$\hat{\mathbf{V}} = W_d^v(W_d^v(\tilde{\mathbf{Y}})) \tag{3}$$
$$\widetilde{\mathbf{Q}} = W_d^{q1}(W_p^{q1}(\tilde{\mathbf{Y}})) \tag{4}$$
$$\widetilde{\mathbf{K}} = \hat{W}_d^{k1}(\hat{W}_p^{k1}(\tilde{\mathbf{X}})) \tag{5}$$
$$\widetilde{\mathbf{V}} = \hat{W}_d^{v1}(\hat{W}_p^{v1}(\tilde{\mathbf{X}})) \tag{6}$$

where $W_p^{(\cdot)}$ and $W_d^{(\cdot)}$ denote the 1×1 pointwise convolution and the 3×3 depthwise convolution, respectively. In the network, this article uses unbiased convolutional layers. Subsequently, we divided $\hat{\mathbf{Q}}$, $\hat{\mathbf{K}}$, $\hat{\mathbf{V}}$, $\widetilde{\mathbf{Q}}$, $\widetilde{\mathbf{K}}$, and $\widetilde{\mathbf{V}}$ into N heads along the spectral channel dimension: $\hat{\mathbf{Q}} = [\hat{\mathbf{Q}}_1, \hat{\mathbf{Q}}_2, \ldots, \hat{\mathbf{Q}}_N]$, $\hat{\mathbf{K}} = [\hat{\mathbf{K}}_1, \hat{\mathbf{K}}_2, \ldots, \hat{\mathbf{K}}_N]$, $\hat{\mathbf{V}} = [\hat{\mathbf{V}}_1, \hat{\mathbf{V}}_2, \ldots, \hat{\mathbf{V}}_N]$, $\widetilde{\mathbf{Q}} = [\widetilde{\mathbf{Q}}_1, \widetilde{\mathbf{Q}}_2, \ldots, \widetilde{\mathbf{Q}}_N]$, $\widetilde{\mathbf{K}} = [\widetilde{\mathbf{K}}_1, \widetilde{\mathbf{K}}_2, \ldots, \widetilde{\mathbf{K}}_N]$, and $\widetilde{\mathbf{V}} = [\widetilde{\mathbf{V}}_1, \widetilde{\mathbf{V}}_2, \ldots, \widetilde{\mathbf{V}}_N]$. The number of channels in each head is $d_h = (C/N)$. Please note that some details have been omitted for simplicity. As shown in Fig. 3(b) and (d), which depict the case of $N = 1$.

2) Multihead Feature Map Attention: As shown in Fig. 3(d), we first deform the query and key to generate a transpose space self-attentive map of size using dot product interactions. Then, MFMSA calculates the self-attention for each $\hat{h}_j$

$$\hat{\mathbf{A}}_j = \text{softmax}(\hat{\mathbf{Q}}_j \hat{\mathbf{K}}_j^T / \sqrt{\alpha_j}) \tag{7}$$
$$\hat{h}_j = \hat{\mathbf{A}}_j \hat{\mathbf{V}}_j \tag{8}$$

where α_j represents the dimension of $\hat{\mathbf{K}}_j$ for scaling and $(.)^T$ is the transpose operator. Then, projections are made by concatenating the outputs of the N heads and then adding the positional embedding

$$\text{MFMA}(\tilde{\mathbf{X}}, \tilde{\mathbf{Y}}) = WD_1(\text{cat}(\hat{h}_j)_{j=1}^N) \tag{9}$$

where cat$(\cdot)$ represents the concatenation operator.

3) Multihead Feature Channel Attention: As shown in Fig. 3(b), our MFCA treats each channel representation as a marker and calculates the self-concentration of $\widetilde{h}_j$

$$\widetilde{\mathbf{A}}_j = \text{softmax}(\beta_j \widetilde{\mathbf{K}}_j^T \widetilde{\mathbf{Q}}_j) \tag{10}$$
$$\widetilde{h}_j = \widetilde{\mathbf{A}}_j \widetilde{\mathbf{V}}_j. \tag{11}$$

We use the learnable parameter β_j to adjust the self-attentive matrix $\widetilde{\mathbf{A}}_j$ by reweighting the matrix multiplication $\widetilde{\mathbf{Q}}_j \widetilde{\mathbf{K}}_j^T$ within $\widetilde{h}_j$. The MFCA can, therefore, be expressed as

$$\text{MFCA}(\tilde{\mathbf{X}}, \tilde{\mathbf{Y}}) = WD_1(\text{cat}(\widetilde{h}_j)_{j=1}^N). \tag{12}$$

4) Multiscale Convolutional Gated FN: To transform the features, a regular FN [50], [53] operates separately and identically on each pixel position. It uses two 1×1 convolutions: one extending the feature channel and the other reducing

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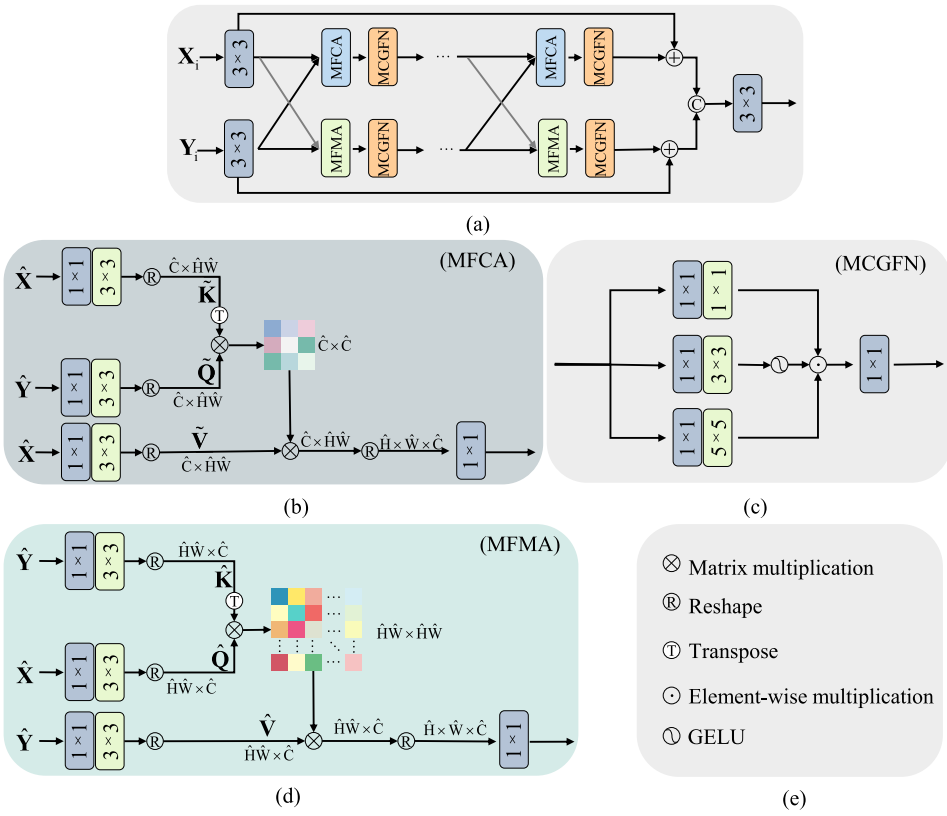


Fig. 3. This study proposes two attention mechanisms, namely, (a) MTB, (b) MFMA, (c) MFCA, (d) MCGFN, and (e) interpretation section of the icon. MFMA is characterized by the use of query ($\hat{\mathbf{Q}}$), key ($\hat{\mathbf{K}}$), and value ($\hat{\mathbf{V}}$) components to enable targeted attention at specific regions of the feature map. Conversely, MFCA leverages query ($\widetilde{\mathbf{Q}}$), key ($\widetilde{\mathbf{K}}$), and value ($\widetilde{\mathbf{V}}$) parameters to facilitate channel-level attention. The proposed attention mechanisms have the potential to improve feature selection and enhance the modeling of spatial and spectral dependencies within the data.

the channel to the original input dimension. The paper [58] proposed using gating mechanisms and deep convolution to construct FNs. In Fig. 3(c), we propose the use of convolutions at different scales to extract image features. Elementwise multiplication is employed on these features to encode information from the spatially adjacent pixel locations, facilitating the learning of effective recovery of local image structures. Given an input tensor $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$, MCGFN is formulated as

$$\text{MCGFN}(\mathbf{I}) = WP_1(\text{MGating}(\mathbf{I})) \tag{13}$$

$$\text{MGating}(\mathbf{I}) = \Psi(WD_3(WP_1(\mathbf{I}))) \odot WD_1(WP_1(\mathbf{I})) \odot WD_5(WP_1(\mathbf{I})) \tag{14}$$

where $\odot$ denotes the element multiplication and Ψ denotes the Gaussian error linear unit (GELU) nonlinearity. In summary, MCGFN controls the flow of information through the layers in our pipeline, thus allowing each layer to focus on fine details complementary to the other layers.

We first extract the input HSI and MSI features using an MTB, as shown in Fig. 3. The module consists of two parts one is to mine the spatial information of the image to complement the HSI using a Transformer composed of traditional subattention mechanisms; the other is to mine the spectral information of the image to complement the MSI using a Transformer composed of subattention mechanisms for the channels. The following is a detailed description of these two components:

$$\mathbf{M}_l' = \text{MFCA}(\text{LN}(\mathbf{N}_{l-1}), \text{LN}(\mathbf{M}_{l-1})) + \mathbf{M}_{l-1} \tag{15}$$

$$\mathbf{M}_l = \text{MCGFN}(\text{LN}(\mathbf{M}_l')) + \mathbf{M}_l' \tag{16}$$

$$\mathbf{N}_l' = \text{MFMA}(\text{LN}(\mathbf{N}_{l-1}), \text{LN}(\mathbf{M}_{l-1})) + \mathbf{N}_{l-1} \tag{17}$$

$$\mathbf{N}_l = \text{MCGFN}(\text{LN}(\mathbf{N}_l')) + \mathbf{N}_l' \tag{18}$$

$$\mathbf{R} = \mathbf{M}_L + \hat{\mathbf{Y}}, \mathbf{Q} = \mathbf{N}_L + \hat{\mathbf{X}} \tag{19}$$

where the outputs $\mathbf{M}_{l-1}$ and $\mathbf{N}_{l-1}$ are from the previous phase, $\mathbf{M}_l$ and $\mathbf{N}_l$ represent the results of phase l, $l \in \{1, 2, \ldots, L\}$, $\mathbf{M}_0 = \hat{\mathbf{Y}}$, and $\mathbf{N}_0 = \hat{\mathbf{X}}$. LN$(\cdot)$ denotes the layer-norm layer.

We feed the inputs HSI and MSI image to MTB and then concatenate the results of these two parts together to obtain better spatial-spectral information. We then deploy a 3×3 convolutional layer to aggregate the spatial-spectral information from different domains to enhance our network spatial-spectral representation. This process can be expressed as follows:

$$\text{MTB}(\hat{\mathbf{X}}, \hat{\mathbf{Y}}) = \text{Conv}_{3\times3}(\text{cat}(\mathbf{R}, \mathbf{Q})) \tag{20}$$

where $\text{Conv}_{3\times3}$ represents the convolution operation and its kernel size is 3×3.

B. Multi-Input Encoders

It has been shown that different levels of blurring can be better handled in multiscale images [59], [60]. Various

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CNN-based deblurring methods have adopted this idea, using blurred images at different scales as input to each subnetwork [37], [61], [62]. In our MIMO-SST, not only are features extracted from the encoder block (EB) in addition to the reduced features but also features are extracted from the downsampled blurred images, and then, the two features are combined. By exploiting the complementary information of the reduced features and the features obtained from the downsampled images, our EB is expected to be able to handle HSI super-resolution problems. The use of multiscale images as input to a single UNet has also proved effective in other tasks.

In order to fully mine the fused HSIs, we use the channelwise Transformer block (CTB) to mine the spectral information of the HSIs and the UNet to mine the multiscale spatial information. It consists of the input of Q, K, and V in MFCA changed to fused HSI features and combined with MCGFN as its FN. The input to the process is $\mathbf{A} \in \mathbb{R}^{H\times W\times C}$, and this process can be expressed as follows:

$$\mathbf{P}'_k = \text{MFCA}(\text{LN}(\mathbf{P}_{k-1}), \text{LN}(\mathbf{P}_{k-1})) + \mathbf{P}_{k-1} \tag{21}$$

$$\mathbf{P}_k = \text{MCGFN}(\text{LN}(\mathbf{P}'_k)) + \mathbf{P}'_k \tag{22}$$

$$\mathbf{S} = \mathbf{P}_K + \mathbf{A} \tag{23}$$

where the inputs $\mathbf{P}_{k-1}$ are from the previous phase and $\mathbf{P}_k$ represents the results of phase k, $k \in \{1, 2, \ldots, K\}$; $\mathbf{P}_0 = \mathbf{A}$. Thus, CTB can be expressed as $\text{CTB}(\mathbf{A}) = \mathbf{S}$.

1) Pixelshuffle: The underadoption operation is required in the UNet, and here, we use the Pixelshuffle(·) [63] formula expression as shown in the following. The key idea in Pixelshuffle is to reshape the input $\mathbf{U} \in \mathbb{R}^{H\times W\times Cr^2}$ into a new tensor $\mathbf{V} \in \mathbb{R}^{rH\times rW\times C}$, with r being the upsampling factor. This process is expressed by the following equation:

$$\mathbf{V}_{i,j,c} = \text{PS}(\mathbf{U}) = \mathbf{U}_{\llcorner i/r \lrcorner, \llcorner j/r \lrcorner, Cr\text{mod}(j,r)+C\text{mod}(i,r)+c} \tag{24}$$

where mod is a modulo operation and $\llcorner\cdot\lrcorner$ is a rounding operation. Since this operation reduces the number of channels, we designed a convolution of 3×3 to keep the number of channels constant. PS(·) represents the Pixelshuffle operation. Pixelshuffle can also be replaced by an interpolation operator, performing about the same. Pixelunshuffle is the inverse process of Pixelshuffle.

To begin with, we use CTB to extract features from downsampled multispectral and HSIs. As shown in Fig. 2, after this module and combined with (20), the equation is expressed as follows:

$$E_i = \begin{cases} \text{CTB}(\text{MTB}(\hat{\mathbf{X}}_i, \mathbf{Up}(\hat{\mathbf{Y}}_i))), & i = 1 \\ \text{CTB}(\text{cat}(\text{MTB}(\hat{\mathbf{X}}_i, \mathbf{Up}(\hat{\mathbf{Y}}_i)), \text{PUS}(E_{i-1}))), & i = 2, 3 \end{cases} \tag{25}$$

where $\hat{\mathbf{X}}_1$, $\hat{\mathbf{X}}_2$, and $\hat{\mathbf{X}}_3$ represent the features obtained by applying a 3×3 convolution with a channel size of 48 to the original, twofold, and fourfold downsampled MSIs, respectively; $\hat{\mathbf{Y}}_1$, $\hat{\mathbf{Y}}_2$, and $\hat{\mathbf{Y}}_3$ represent the features obtained by applying a 3×3 convolution with a channel size of 48 to the fourfold, twofold upsampling, and original HSIs, respectively. Here, $\mathbf{Up}(.)$ denotes that the HSI is upsampled by bilinear interpolation to the same spatial size as the MSI. PUS(·) denotes Pixelunshuffle. E_i denotes the output of the encoder at layer i.

C. Multi-Output Decoders

The different decoder blocks (DBs) in MIMO-SST have feature maps of different scales. These multiscale feature maps are believed to be used to model multistacked subnetworks. In contrast to the traditional coarse-to-fine network with intermediate supervision over subnetworks, which we applied to each DB, the image reconstruction at each level and when combined with (20) can be formulated as follows:

$$D_i = \begin{cases} \text{CTB}(E_3), & i = 3 \\ \text{CTB}(\text{cat}(\text{PS}(D_1), E_2)), & i = 2 \\ \text{CTB}(\text{cat}(\text{PS}(D_2), E_1)), & i = 1 \end{cases} \tag{26}$$

where D_i denotes the output of the decoder at layer i. The output of the network can be obtained by passing the D_i through a 3×3 convolution and then adding the residuals to $\mathbf{X}_i$. This process can be expressed as

$$\hat{\mathbf{Z}}_i = \text{Conv}_{3\times 3}(D_i) + \mathbf{X}_i \tag{27}$$

where $\hat{\mathbf{Z}}_i$ denotes the multiscale HSI output by the network at layer i.

D. Loss Function

We also used a multiscale content loss function like other multiscale fusion networks. Using the multiscale content loss function, we found that L1 loss produced better results for our network than mean square error (mse) loss. The content loss Lcont is defined as follows:

$$\mathcal{L}_{\text{cont}} = \sum_{i=1}^{N} \|\mathcal{F}(\mathbf{X}_i, \mathbf{Y}_i; \Theta) - \mathbf{Z}_i\|_1 \tag{28}$$

where $\hat{\mathbf{Z}}_i$ denotes the reconstructed HSI patch. $\mathbf{Z}_1$, $\mathbf{Z}_2$, and $\mathbf{Z}_3$ represent the original, twofold downsampled, and fourfold downsampled HSIs, respectively. $\mathcal{F}(\cdot; \Theta)$ denotes the reconstructed HSI patch by the network with parameters Θ.

Recent research [37] has also suggested that auxiliary loss terms can be used to improve performance in addition to content loss. Auxiliary loss terms that minimize the distance between input and output in the feature space have been widely used in image enhancement and restoration tasks and have shown promising results. As the main objective of fusion is to recover the high-frequency components of an image, this can be achieved by reducing the variance of high-frequency information in HSIs.

We briefly describe the process of extracting high-frequency subband images from the discrete wavelet transform. Images of high-frequency subbands in the three directions are defined as

$$W^1 = \bar{\Phi}\bar{\Psi}(\mathbf{C}) \tag{29}$$

$$W^2 = \bar{\Psi}\bar{\Phi}(\mathbf{C}) \tag{30}$$

$$W^3 = \bar{\Psi}\bar{\Psi}(\mathbf{C}) \tag{31}$$

where $\bar{\Phi}\bar{\Psi}(\mathbf{C})$ represents the convolution of the image $\mathbf{C}$ with the separable filter $\bar{\Phi}\bar{\Psi}$. W^1, W^2, and W^3 represent

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high-frequency subband images in horizontal, vertical, and diagonal directions of the image, respectively.

To this end, we propose a WHF loss function. The WHF loss is used to measure the high-frequency L1 distance between the multiscale ground-truth HSI and the fused HSI as follows:

$$\mathcal{L}_{\mathrm{WHF}} = \sum_{i=1}^{N} \|\mathcal{H}(\mathcal{F}(\mathbf{X}_i, \mathbf{Y}_i; \Theta)) - \mathcal{H}(\mathbf{Z}_i)\|_1 \tag{32}$$

where $\mathcal{H}(\cdot)$ denotes the high-frequency portion extracted by the Haar discrete wavelet transform (filter bank is "DB1"). $\mathcal{H}(\cdot)$ can be expressed as a concatenation of high-frequency information in the horizontal, vertical, and diagonal directions of the input HSI. For the process of $\mathcal{H}(\cdot)$, the specific formula is represented as follows:

$$\mathcal{H}(\mathbf{C}) = \mathrm{cat}(W^1, W^2, W^3). \tag{33}$$

The final loss function for our training network is described in the following:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{cont}} + \lambda \mathcal{L}_{\mathrm{WHF}} \tag{34}$$

where λ is a balancing parameter.

During network training, the adaptive moment estimation (ADAM) optimizer of $\beta_1 = 0.9$ and $\beta_2 = 0.999$ is used with the learning rate $\gamma = 1 \times 10^{-4}$, the number of iterations is 200, and the batch size is 32.

IV. Experiments

In this section, we present in detail the experimental specifics and network performance evaluation of a network consisting of MTB, CTB, MIMO, and WHF modules to address the hyperspectral super-resolution problem. First, the datasets and evaluation metrics used in the experiments are presented. Then, we compare the performance of the proposed MIMO-SST network with state-of-the-art hyperspectral fusion methods. We compare MIMO-SST with three classical methods (coupled nonnegative matrix factorization (CNMF) [17],[1] the subspace regularized method (HySure) [19],[2] and coupled spectral unmixing (CSU) [23][3]) and four state-of-the-art methods (existing techniques) (deep HSI sharpening method (DHSIS) [26],[4] deep blind iterative fusion network (DBIN) [30],[5] a dual U-Net (D-UNet) [46], and Fusformer [34][6]). Finally, an ablation experiment is described to compare the effectiveness of networks with and without additional modules. The experiment is conducted on an NVIDIA 2080TI RTX computer using PyTorch.

A. Experimental Dataset

The performance of the proposed method is validated using three publicly simulated hyperspectral imaging datasets, i.e., the Columbia computer vision laboratory (CAVE) dataset,[7] the Interdisciplinary Computational VisionLab (ICVL) dataset,[8] and the Chikusei dataset.[9] For all three datasets, LR-HSI was obtained by resampling every r pixels downward in the vertical and horizontal directions of the reference image using a Gaussian filter (the mean value is 0 and the standard deviation is 2), i.e., an extraction factor of $r \times r$. At the same time, the observed images from these datasets are used as reference images. We performed spatial enhancements of factors 8, 16, and 32 in the experiments. Partial images of the test set for these three datasets are shown in Fig. 4.

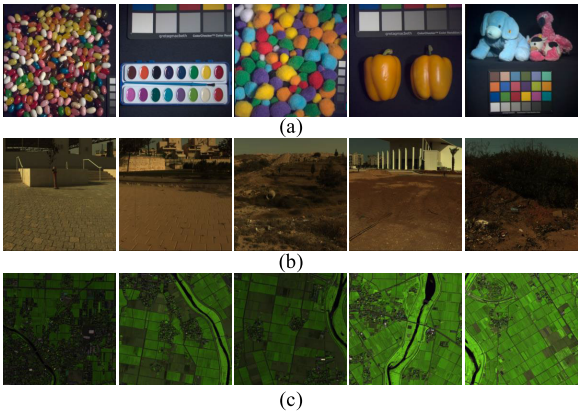


Fig. 4. Examples of the testing images selected from (a) CAVE, (b) ICVL, and (c) Chikusei datasets.

1) CAVE: This dataset has a spatial size of 512 × 512, a band range of 400–700 nm, a wavelength interval of 10 nm, and 31 spectral bands consisting of 32 indoor HSIs. This same situation is simulated by spectrally downsampling the HR-HSI using a subspectral sampling matrix $\mathbf{R}$, where $\mathbf{R}$ used the response function of the Nikon D700 camera.[10] For this dataset, the first 20 HSIs are used for training, and the last 12 HSIs are used for testing.

2) ICVL: There are 31 spectral bands in the dataset, including 201 HSIs, with a spatial size of 1300 × 1392, a band range of 400–700 nm, and a wavelength interval of 10 nm. For convenience, in our experiments, we cropped 1024 pixels from the top left corner of the dataset for training and testing the proposed method. Fifty datasets are selected from 201 datasets for the experiments. In our experiments, the first 30 HSIs are used as training datasets, and the last 20 HSIs are used as test datasets. With the large amount of data required for DL as a training set, these training HSI blocks are used as the training set to train the proposed network.

3) Chikusei: It contains aerial HSI taken by visible and near-infrared (NIR) imaging sensors in the agricultural and urban areas of Tsukiji, Ibaraki, Japan. There are 128 bands in the spectral range from 363 to 1018 nm in this hyperspectral dataset, and the scenes consist of 2517 × 2335 pixels. When black edges in the spatial domain were removed, the central 2048 × 2048 pixels were cropped and extracted for use in our experiments. An IKONOS satellite spectral response function R[11] is used to generate the HR-MSI for this dataset, taking into account the diversity of hyperspectral sensors.

[1] http://naotoyokoya.com/Download.html
[2] https://github.com/alfaiate/HySure
[3] https://github.com/lanha/SupResPALM
[4] https://github.com/renweidian/DHSIS
[5] https://github.com/renweidian/CNN-FUS
[6] https://github.com/J-FHu/Fusformer
[7] http://www.cs.columbia.edu/CAVE/databases/multispectral/
[8] http://icvl.cs.bgu.ac.il/hyperspectral/
[9] http://naotoyokoya.com/Download.html
[10] https://www.maxmax.com/spectral_response.htm
[11] https://www.satimagingcorp.com/satellite-sensors/ikonos/

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TABLE I

CORRESPONDING SIZES OF LR-HSI, HR-MSI, AND HR-HSI FOR THE DIFFERENT DATASETS AT DIFFERENT UPSCALING FACTORS

dataset	Upscaling factor	LR-HSI	HR-MSI	HR-HSI
CAVE and ICVL	8	$4 \times 4 \times 31$	$32 \times 32 \times 3$	$32 \times 32 \times 31$
	16	$2 \times 2 \times 31$	$32 \times 32 \times 3$	$32 \times 32 \times 31$
	32	$1 \times 1 \times 31$	$32 \times 32 \times 3$	$32 \times 32 \times 31$
Chikusei	8	$4 \times 4 \times 128$	$32 \times 32 \times 4$	$32 \times 32 \times 128$
	16	$4 \times 4 \times 128$	$32 \times 32 \times 4$	$32 \times 32 \times 128$
	32	$4 \times 4 \times 128$	$32 \times 32 \times 4$	$32 \times 32 \times 128$

TABLE II

AVERAGE MPSNR, RMSE, ERGAS, SAM, UIQI, AND MSSIM RESULTS OF THE PROPOSED AND COMPARISON METHODS ON THE CAVE DATASET WITH GAUSSIAN BLUR KERNEL AND SCALING FACTORS OF 8, 16, AND 32. THE ↑ INDICATES THAT THE LARGER THE VALUE, THE BETTER THE PERFORMANCE, AND THE ↓ INDICATES THAT THE SMALLER THE VALUE, THE BETTER THE PERFORMANCE. THE BEST AND SECOND-BEST RESULTS ARE **HIGHLIGHTED** AND UNDERLINED, RESPECTIVELY

Scale factor	Indices	CNMF [17]	HySure [19]	CSU [23]	DHSIS [26]	DBIN [30]	D-UNet[46]	Fusformer [34]	MSST [35]	PSRT [36]	MIMO-SST
s=8	MPSNR↑	34.3027	34.7822	35.6888	46.2977	45.7831	46.2823	46.4546	45.3346	46.4069	**47.3029**
	RMSE↓	5.4723	5.3232	4.7694	1.4653	1.5233	1.4759	1.4532	1.5691	1.5030	**1.2942**
	ERGAS↓	2.6006	2.4181	2.2064	0.6641	0.6781	0.6655	0.6511	0.7056	0.6506	**0.5838**
	SAM↓	7.8920	11.5451	7.8787	3.8452	3.6035	4.3319	3.7566	4.4312	**3.5399**	3.6227
	UIQI↑	0.771	0.8043	0.7986	0.9242	0.9260	0.9128	0.9276	0.8928	0.9210	**0.9295**
	MSSIM↑	0.9388	0.9107	0.9531	0.9904	0.9925	0.9885	0.9921	0.9896	0.9924	**0.9929**
s=16	MPSNR↑	26.4938	28.5438	30.9482	41.4026	42.1841	41.5954	42.9389	39.6067	43.2958	**44.5365**
	RMSE↓	13.2830	11.3403	7.9381	2.7171	2.4506	2.5926	2.2401	3.3432	2.2852	**1.9118**
	ERGAS↓	3.0507	2.5827	1.8842	**0.5774**	1.0443	1.1361	0.9514	1.3847	0.9404	0.8104
	SAM↓	11.5677	18.5054	11.4525	5.8874	**4.6401**	6.3317	5.0199	8.9669	4.7308	4.6807
	UIQI↑	0.5742	0.6894	0.6904	0.8859	0.9111	0.8792	0.9075	0.8500	0.9151	**0.9163**
	MSSIM↑	0.8326	0.7907	0.9042	0.9818	0.9889	0.9784	0.9871	0.9757	0.9888	**0.9897**
s=32	MPSNR↑	21.4638	21.0979	26.6491	38.6731	37.6475	36.7921	35.5739	37.2190	40.0611	**40.6770**
	RMSE↓	24.7266	26.2165	12.8841	3.8456	4.7977	4.4249	5.7827	4.3533	3.3108	**3.0982**
	ERGAS↓	2.5815	3.0317	1.4651	**0.3939**	1.8572	1.9562	2.3029	1.7709	1.3297	1.2726
	SAM↓	18.5055	30.9460	14.3466	8.2859	**7.1367**	10.3624	12.3454	10.2359	7.5237	7.2175
	UIQI↑	0.3848	0.5259	0.5719	0.8484	0.8692	0.8030	0.8165	0.8122	0.8805	**0.8837**
	MSSIM↑	0.6785	0.6487	0.8486	0.9728	0.9769	0.9454	0.9431	0.9607	0.9710	**0.9807**

TABLE III

AVERAGE MPSNR, RMSE, ERGAS, SAM, UIQI, AND MSSIM RESULTS OF THE PROPOSED AND COMPARISON METHODS ON THE ICVL DATASET WITH GAUSSIAN BLUR KERNEL AND SCALING FACTORS OF 8, 16, AND 32. THE ↑ INDICATES THAT THE LARGER THE VALUE, THE BETTER THE PERFORMANCE, AND THE ↓ INDICATES THAT THE SMALLER THE VALUE, THE BETTER THE PERFORMANCE. THE BEST AND SECOND-BEST RESULTS ARE **HIGHLIGHTED** AND UNDERLINED, RESPECTIVELY

Scale factor	Indices	CNMF [17]	HySure [19]	CSU [23]	DHSIS [26]	DBIN [30]	D-UNet[46]	Fusformer [34]	MSST [35]	PSRT [36]	MIMO-SST
s=8	MPSNR↑	42.7429	41.0796	41.5650	49.7071	50.3109	52.0298	51.5171	47.3504	51.8818	**52.3451**
	RMSE↓	2.0934	2.5573	2.3981	0.9146	0.8549	0.6855	0.7597	1.2517	0.7048	**0.6596**
	ERGAS↓	0.7413	1.0102	0.8612	0.4968	0.4413	0.3486	0.4195	0.4607	0.3499	**0.3334**
	SAM↓	1.7214	2.7047	2.0628	1.3959	1.2836	1.1609	**1.0954**	1.6699	1.1268	1.1267
	UIQI↑	0.9125	0.9243	0.9240	0.9485	0.9536	0.9593	0.9591	0.9544	0.9604	**0.9591**
	MSSIM↑	0.9838	0.9842	0.9873	0.9942	0.9957	0.9959	0.9959	0.9956	0.9960	**0.9962**
s=16	MPSNR↑	36.5197	35.4180	36.6287	47.9687	48.9072	50.3195	50.0437	44.5512	50.3234	**51.3090**
	RMSE↓	4.5278	5.1881	4.3334	1.1802	1.0498	0.8799	0.9031	1.7906	0.8853	**0.7781**
	ERGAS↓	0.7000	0.7966	0.7030	**0.3120**	0.5424	0.4356	0.4454	0.6862	0.4286	0.3997
	SAM↓	2.5573	5.3196	2.8482	1.6989	1.5081	1.3957	1.2918	2.4739	1.3011	**1.2411**
	UIQI↑	0.8295	0.8888	0.8715	0.9400	0.9461	0.9527	0.9551	0.9267	0.9551	**0.9570**
	MSSIM↑	0.9546	0.9536	0.9723	0.9931	0.9945	0.9953	0.9954	0.9917	0.9956	**0.9960**
s=32	MPSNR↑	30.9816	30.7356	33.0675	43.6325	38.2204	48.6118	47.5920	46.3043	49.1295	**49.2800**
	RMSE↓	8.7894	9.1533	6.6887	1.9399	5.6025	1.1271	1.5012	1.4478	1.0563	**1.0384**
	ERGAS↓	0.6550	0.6892	0.5261	0.2862	2.1582	0.5252	**0.2329**	0.6476	0.4714	0.4806
	SAM↓	3.7144	9.5154	3.5467	2.8897	6.9125	1.6788	1.6641	2.0199	1.7586	**1.6127**
	UIQI↑	0.7040	0.8234	0.7951	0.9180	0.8561	0.9475	0.9474	0.9282	0.9416	**0.9500**
	MSSIM↑	0.9044	0.8980	0.9517	0.9872	0.9713	0.9948	0.9938	0.9924	**0.9951**	**0.9951**

The information regarding the magnitudes of LR-HSI, HR-MSI, and HR-HSI for different datasets across varying upsampling factors is presented in Table I. These data serve to provide valuable insights into the effects of upscaling on the dimensions of the resulting image data.

B. Quantitative Indicators

Several different spatial and spectral quality measures are used to assess the quality of the proposed HSI fusion method. In accordance with the literature [64], [65], [66], [67], [68], [69], we use the mean PSNR (MPSNR), spectral

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TABLE IV

AVERAGE MPSNR, RMSE, ERGAS, SAM, UIQI, AND MSSIM RESULTS OF THE PROPOSED AND COMPARISON METHODS ON THE CHIKUSEI DATASET WITH GAUSSIAN BLUR KERNEL AND SCALING FACTORS OF 8, 16, AND 32. THE ↑ INDICATES THAT THE LARGER THE VALUE, THE BETTER THE PERFORMANCE, AND THE ↓ INDICATES THAT THE SMALLER THE VALUE, THE BETTER THE PERFORMANCE. THE BEST AND SECOND-BEST RESULTS ARE **HIGHLIGHTED** AND UNDERLINED, RESPECTIVELY

Scale factor	Indices	CNMF [17]	HySure [19]	CSU [23]	DHSIS [26]	DBIN [30]	D-UNet[46]	Fusformer [34]	MSST [35]	PSRT [36]	MIMO-SST
s=8	MPSNR↑	23.2415	21.5400	23.2267	27.1123	29.8938	29.9550	29.3536	30.6607	29.1437	**30.9155**
	RMSE↓	27.5074	30.0995	26.8957	21.2084	12.2755	11.7224	14.6929	11.5783	13.8759	**11.2684**
	ERGAS↓	11.4413	14.0526	11.5262	8.1020	4.8705	4.9811	6.3320	4.3550	5.6833	**2.2080**
	SAM↓	4.2121	6.5111	3.3637	16.7390	3.5885	4.6340	2.4021	2.2294	**2.1384**	2.3513
	UIQI↑	0.6871	0.5923	0.7042	0.6566	0.8680	0.8598	0.8517	0.8854	0.8061	**0.9007**
	MSSIM↑	0.7792	0.6993	0.7831	0.6819	0.9167	0.9122	0.9001	0.9340	0.8939	**0.9403**
s=16	MPSNR↑	19.8532	17.6746	21.6445	25.5592	26.9923	28.0212	27.3656	28.7094	27.1363	**30.1919**
	RMSE↓	36.2786	44.7360	31.9213	21.7916	17.6094	14.9517	16.1168	15.3571	16.4177	**12.1865**
	ERGAS↓	16.0684	10.6288	13.5389	4.2079	6.6532	5.9951	**3.3661**	5.4754	6.8172	4.7301
	SAM↓	7.6024	12.4582	4.8723	18.8792	**2.6261**	6.3858	3.6060	3.7555	3.6880	2.6720
	UIQI↑	0.4261	0.4361	0.6118	0.5055	0.8101	0.7614	0.8109	0.8377	0.7388	**0.8855**
	MSSIM↑	0.6104	0.5546	0.6965	0.6632	0.8929	0.8279	0.8880	0.9050	0.8597	**0.9323**
s=32	MPSNR↑	17.4211	14.3532	20.3353	25.2770	26.2270	27.5868	23.0661	27.7417	27.7332	**29.6724**
	RMSE↓	46.3714	64.5680	37.5688	22.8627	17.6542	15.5390	24.6118	16.6585	15.2161	**12.6238**
	ERGAS↓	5.3359	7.7948	3.9384	**2.2812**	7.6213	6.3360	6.1665	6.1800	5.9058	2.5286
	SAM↓	10.7349	18.6708	5.9179	21.9690	7.0000	6.8453	6.3871	3.9256	3.8157	**2.7053**
	UIQI↑	0.2755	0.3313	0.5402	0.5140	0.6768	0.7382	0.6944	0.8046	0.8079	**0.8776**
	MSSIM↑	0.4981	0.4366	0.6342	0.6884	0.7742	0.8104	0.7982	0.8758	0.8949	**0.9272**

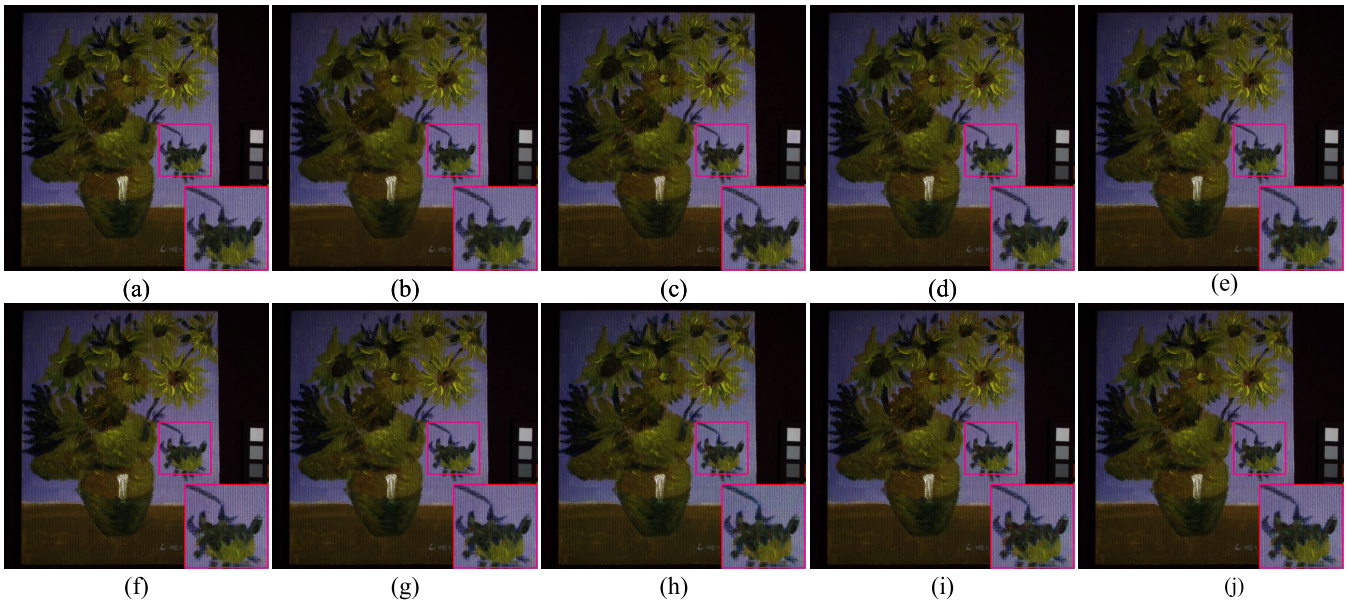


Fig. 5. Fused images of oil painting selected from the CAVE dataset with pseudocolor composite map (bands: 30, 20, and 10). (a) CNMF [17]. (b) HySure [19]. (c) CSU [23]. (d) DHSIS [26]. (e) DBIN [30]. (f) D-UNet [46]. (g) Fusformer [34]. (h) MSST [35]. (i) PSRT [36]. (j) MIMO-SST.

angle mapping (SAM), mean structural similarity indicator (MSSIM), erreur relative globale adimensionnelle de synthese (ERGAS), root mean square error (RMSE), and universal image quality index (UIQI). The extensive HSI handling community has used the metrics for a wide range of applications to the assessment of the spectral and spatial resolutions of the fusion.

C. Experimental Results

Tables II–IV show the evaluation results of the different methods. For the CAVE, ICVL, and Chikusei datasets, different fusion methods are first evaluated on 12, 20, and eight test datasets, respectively, and then, the mean of the evaluation metrics is calculated. According to Tables II–IV, the proposed method achieves higher MPSNR, UIQI, and MSSIM metrics and lower RMSE, ERGAS, and SAM metrics; this means that the HSI reconstructed by the proposed method has a better spatial structure and less spectral distortion than the comparison methods.

Due to the fused HR-HSIs shown in Figs. 5, 6, and 7 being close to each other, the visual heat maps of mean squared error images are depicted in Figs. 8, 9, and 10 to visually highlight their differences, where the blue color indicates small errors and the red color indicates significant errors.

1) Results on the CAVE Dataset: Based on Fig. 8, it is evident that CNMF, HySure, CSU, and DHSIS fusion methods exhibit significant reconstruction errors in the squares of oil paintings and along the edges of sunflower petals.

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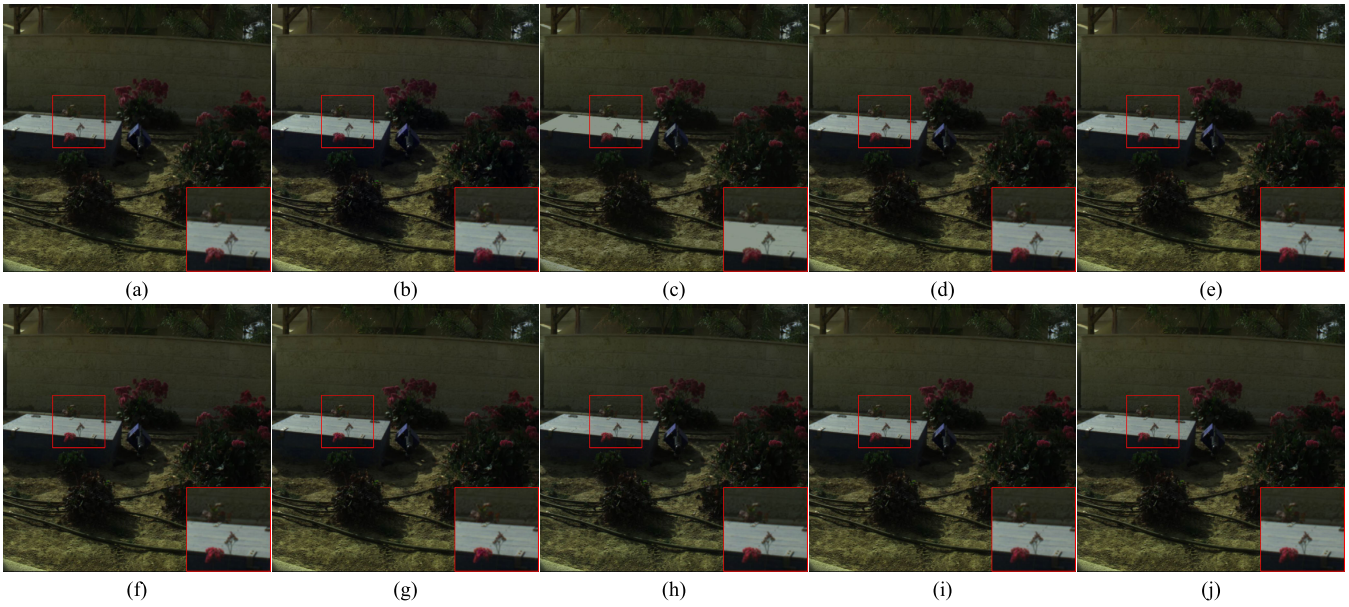


Fig. 6. Composite pseudocolor map of the fused images of eve_0331-1618 selected from the ICVl dataset (bands: 30, 20, and 10). (a) CNMF [17]. (b) HySure [19]. (c) CSU [23]. (d) DHSIS [26]. (e) DBIN [30]. (f) D-UNet [46]. (g) Fusformer [34]. (h) MSST [35]. (i) PSRT [36]. (j) MIMO-SST.

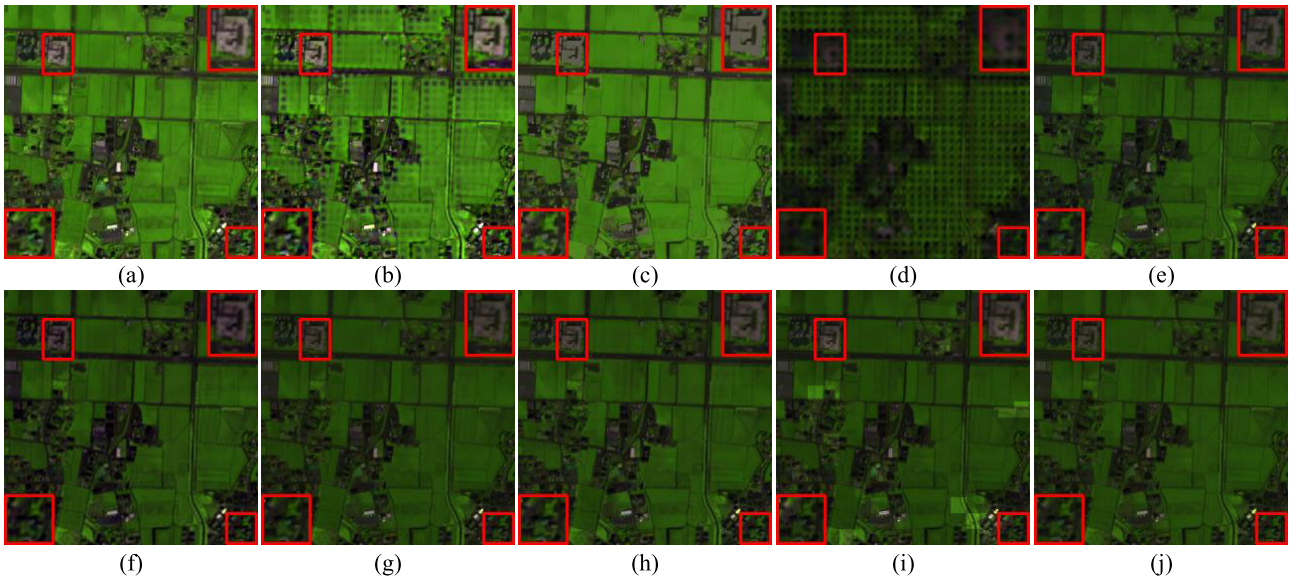


Fig. 7. Composite pseudocolor map of the fused images of Chikusei (bands: 70, 100, and 36). (a) CNMF [17]. (b) HySure [19]. (c) CSU [23]. (d) DHSIS [26]. (e) DBIN [30]. (f) D-UNet [46]. (g) Fusformer [34]. (h) MSST [35]. (i) PSRT [36]. (j) MIMO-SST.

The reconstructed images by DBIN, D-UNet, and MSST methods display pronounced oil painting texture artifacts, resulting in more severe reconstruction errors. The Fusformer and PSRT methods exhibit notably higher reconstruction errors, particularly in the case of sunflower petals. Given that this article employs high-frequency detail extraction as a loss function, our method holds an advantage over others in terms of reconstructing image textures. Comparing reconstruction errors, our proposed method minimizes these errors compared to other approaches. Per-band PSNR and SSIM values for the reference and fused images are presented in Figs. 11(a) and 12(a), respectively. These figures indicate that the proposed method achieves the highest indices on each band, highlighting its superior performance with minimal reconstruction errors. In addition, Fig. 13(a) illustrates the spectral angle mapper (SAM) values of the reference and fused images across each band.

2) Results on the ICVL Dataset: In reference to Fig. 9, it is evident that CNMF, HySure, and CSU fusion methods exhibit significant reconstruction errors in the flowers adjacent to the visor and those on the visor itself. DHSIS, Fusformer, and PSRT methods display larger reconstruction errors in the flowers adjacent to the visor and other floral regions. DBIN and D-UNet methods demonstrate relatively larger reconstruction errors on the side of the visor. Small frames in the MSST method may incorrectly style the extracted texture information, leading to errors. From the figure, it is evident that the proposed MIMO-SST method minimizes reconstruction errors

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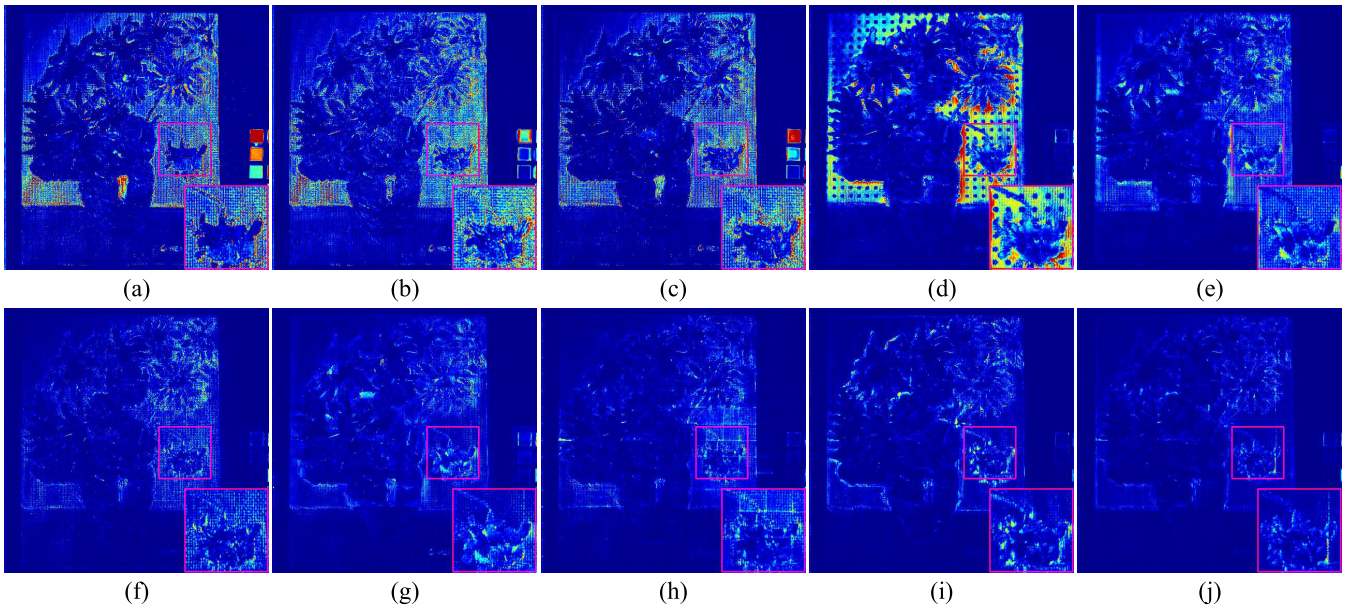


Fig. 8. Comparison methods of fused images of oil painting with mean square error images. (a) CNMF [17]. (b) HySure [19]. (c) CSU [23]. (d) DHSIS [26]. (e) DBIN [30]. (f) D-UNet [46]. (g) Fusformer [34]. (h) MSST [35]. (i) PSRT [36]. (j) MIMO-SST.

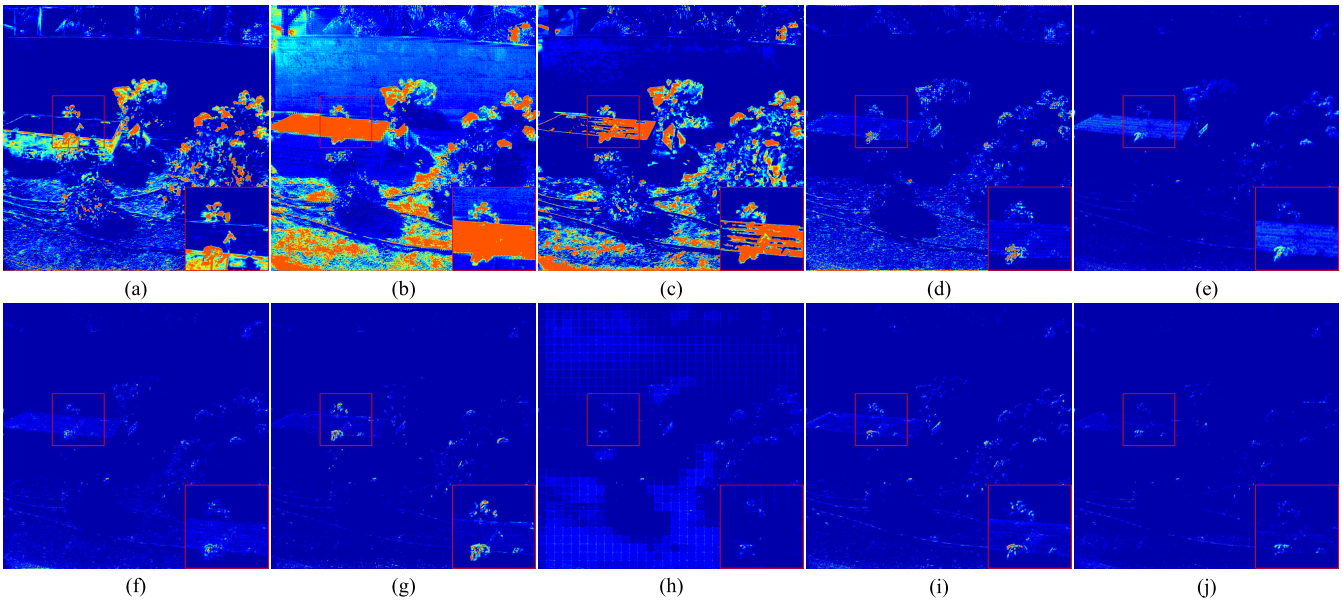


Fig. 9. Average error image corresponding to the fusion images comparison methods of rsh_0406-1356. (a) CNMF [17]. (b) HySure [19]. (c) CSU [23]. (d) DHSIS [26]. (e) DBIN [30]. (f) D-UNet [46]. (g) Fusformer [34]. (h) MSST [35]. (i) PSRT [36]. (j) MIMO-SST.

effectively. The utilization of Transformer-based methods is advantageous for reconstructing larger objects, such as sun visors, due to their ability to capture long-range dependencies. By employing Transformer-based architecture and detail extraction loss, this article's method excels in recovering both larger objects and finer details. The MPSNR and MSSIM values for each band of the reference and fused images are depicted in Figs. 11(b) and 12(b), respectively. These figures indicate that the proposed method achieves the highest indices in each band, signifying its small reconstruction error. Furthermore, Fig. 13(b) illustrates the SAM values for the reference and fused images across each band. Once again, the proposed method demonstrates the lowest indices, indicating reduced spectral distortion.

3) Results on the Chikusei Dataset: From Fig. 10, it is clear that the fusion results of CNMF, HySure, and CSU exhibit significant reconstruction errors in specific areas, such as houses and ridges. Moreover, the images reconstructed by the DHSIS method display reconstruction errors caused by speckle noise. In addition, the DBIN, D-UNet, Fusformer, MSST, and PSRT methods demonstrate significant reconstruction errors when reconstructing houses. Conversely, the proposed method described in this article aims to minimize the reconstruction error. This proposed method follows a supervised approach with distinct scales of outputs, thereby possessing an advantage in reconstructing larger objects. To quantitatively evaluate the performance, the PSNR and the structural similarity index (SSIM) of each band of the reference and fused images

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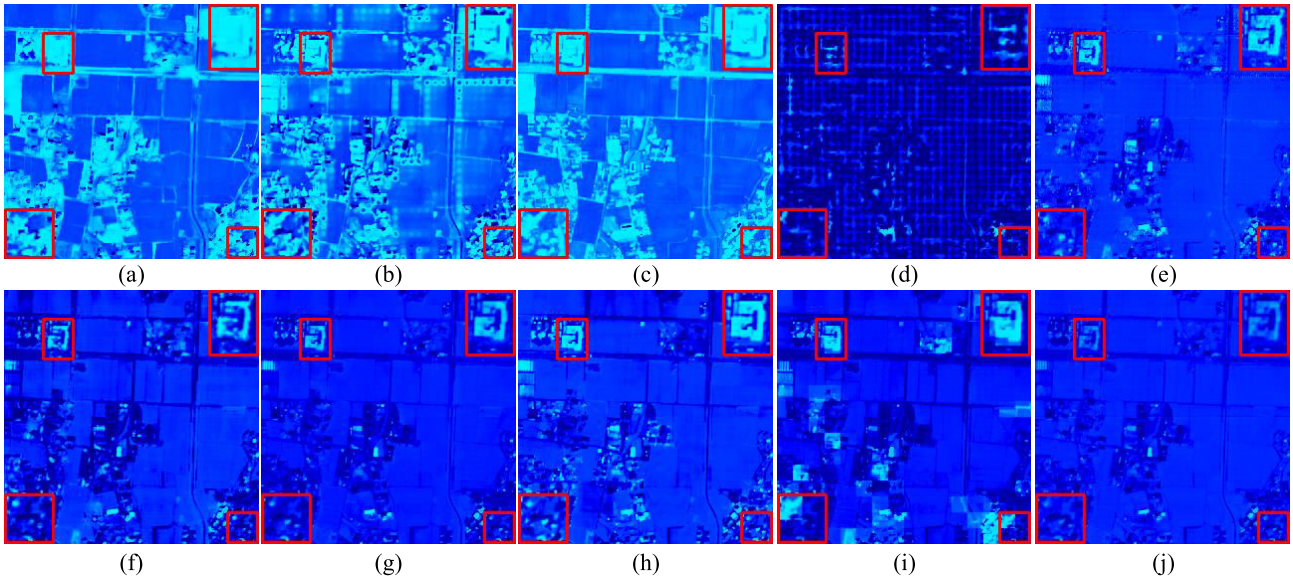


Fig. 10. Average error image corresponding to the fusion images comparison methods of the Chikusei dataset. (a) CNMF [17]. (b) HySure [19]. (c) CSU [23]. (d) DHSIS [26]. (e) DBIN [30]. (f) D-UNet [46]. (g) Fusformer [34]. (h) MSST [35]. (i) PSRT [36]. (j) MIMO-SST.

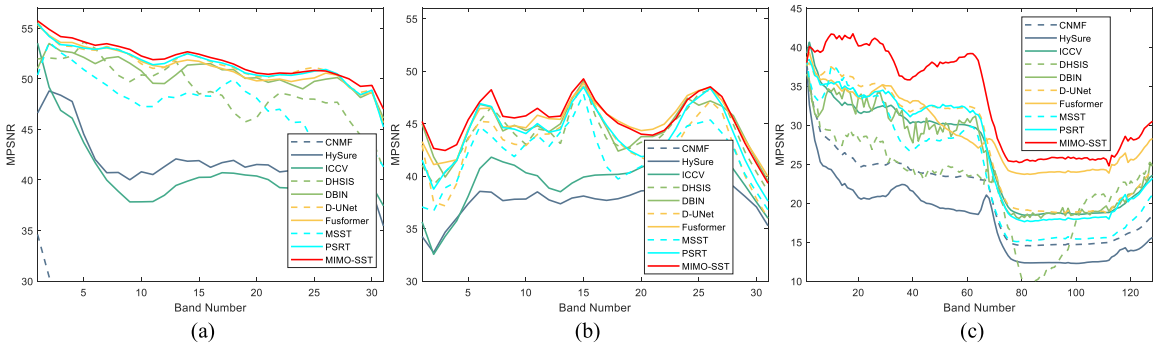


Fig. 11. Comparison of MPSNR curves in different bands. (a) Real and fake apples of the CAVE datasets. (b) rsh_0406-1356 of the ICVL datasets. (c) Chikusei datasets.

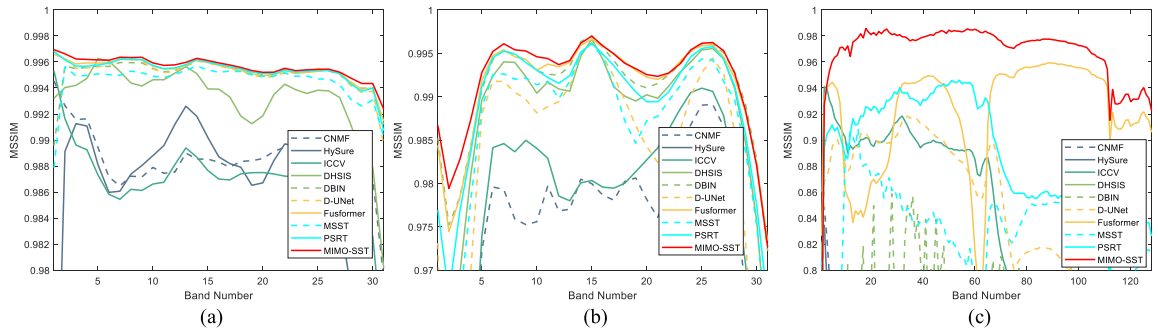


Fig. 12. Comparison of MSSIM curves for each band. (a) Real and fake apples of the CAVE datasets. (b) rsh_0406-1356 of the ICVL datasets. (c) Chikusei datasets.

are presented in Figs. 11(c) and 12(c), respectively. From these plots, it can be observed that our proposed method achieves the highest values in most bands, indicating that the reconstruction error of the method discussed in this article is relatively small. In addition, the SAM for each band of the reference and fused images is depicted in Fig. 13(c). According to the figure, it is apparent that our method yields the lowest indices in each band, suggesting that the proposed method in this article exhibits reduced spectral distortion.

4) Results on Different Noise Levels: In the fusion task, MSI and HSI are often affected by noise [70], so noise is added to the testing images to investigate the proposed method's robustness further. While we simulated HSI and MSI from HR-HSI, Gaussian noise was added to HSI and MSI with signal-to-noise ratios (SNRs) ranging from 10 and 20 to 30 dB. The metrics evaluated for the CAVE dataset are calculated for each noise level and then averaged, as shown in Table V. It is evident from the table that the proposed MIMO-SST method shows more advantages when the SNR is 10, 20, and 30 dB.

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TABLE V

QUANTITATIVE EVALUATION OF VARIOUS ALGORITHMS IS CONDUCTED ON THE CAVE DATASET THROUGH THE ADDITION OF DIVERSE FORMS OF NOISE

Noise level	Indices	CNMF [17]	HySure [19]	CSU [23]	DHSIS [26]	DBIN [30]	D-UNet[46]	Fusformer [34]	MSST [35]	PSRT [36]	MIMO-SST
SNR=10	MPSNR↑	22.8397	22.3578	26.4966	21.9532	35.0170	28.6991	33.9451	30.3120	33.6144	**36.3752**
	RMSE↓	21.3987	23.4086	13.5162	22.8121	5.2609	10.0710	5.8155	9.4400	6.2920	**4.2775**
	ERGAS↓	8.2492	8.8265	5.3823	9.3475	2.1135	4.2946	2.3087	3.8930	2.5768	**1.8406**
	SAM↓	34.3244	35.3193	21.5722	37.5199	**9.9285**	21.5142	11.6897	17.5742	16.1412	11.3313
	UIQI↑	0.3590	0.3500	0.4542	0.3407	0.6739	0.4967	0.6319	0.5881	0.6233	**0.7237**
	MSSIM↑	0.3249	0.3084	0.4746	0.2977	0.9150	0.6173	0.8926	0.8341	0.8735	**0.9269**
SNR=20	MPSNR↑	31.9185	31.0712	32.9274	30.9609	38.1445	37.1938	39.0616	37.1845	38.6256	**40.4362**
	RMSE↓	7.4887	8.0058	6.1999	8.0319	3.5779	3.8873	3.2371	4.0757	3.4756	**2.6948**
	ERGAS↓	3.0593	3.2952	2.7268	3.3119	1.4923	1.6181	1.3209	1.6823	1.4300	**1.1834**
	SAM↓	18.5690	19.4008	13.1142	23.0604	7.8114	10.4399	**7.2201**	9.8062	8.7668	8.6127
	UIQI↑	0.5777	0.5856	0.6228	0.5516	0.7547	0.6929	0.7444	0.7368	0.7364	**0.7933**
	MSSIM↑	0.7340	0.7313	0.8187	0.6774	0.9557	0.9150	0.9596	0.9509	0.9545	**0.9649**
SNR=30	MPSNR↑	36.7218	34.3774	35.2222	35.4432	39.8444	41.0836	42.7465	40.9385	42.8228	**44.4637**
	RMSE↓	4.1746	5.4637	4.7429	4.8047	3.2586	2.4893	2.1088	2.6586	2.1088	**1.7018**
	ERGAS↓	1.9032	2.4352	2.2738	2.2500	1.4018	1.0720	0.8854	1.1432	0.8871	**0.7671**
	SAM↓	11.0044	12.2398	10.0157	15.8815	6.5077	7.5541	6.3011	7.6591	6.0856	**5.2467**
	UIQI↑	0.7157	0.7326	0.7134	0.6575	0.8217	0.7848	0.8253	0.8137	0.8212	**0.8639**
	MSSIM↑	0.9155	0.8972	0.9259	0.8653	0.9745	0.9613	0.9781	0.9739	0.9799	**0.9860**

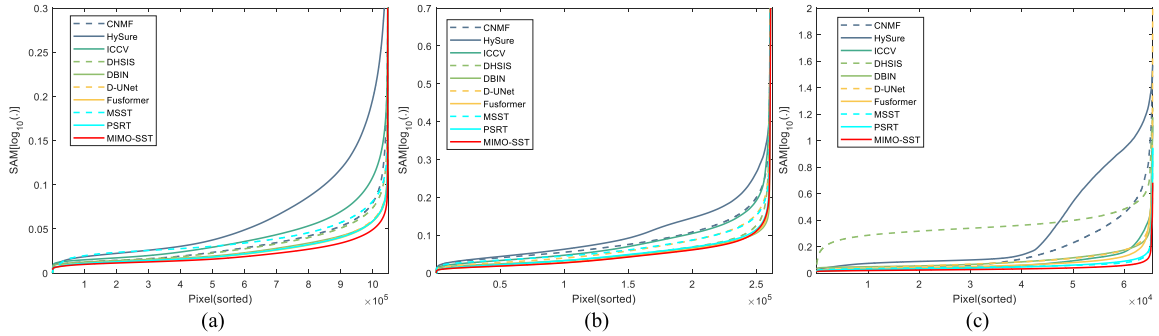


Fig. 13. Corresponding SAM curves of different methods are compared. (a) Real and fake apples of the CAVE datasets. (b) rsh_0406-1356 of the ICVL datasets. (c) Chikusei datasets.

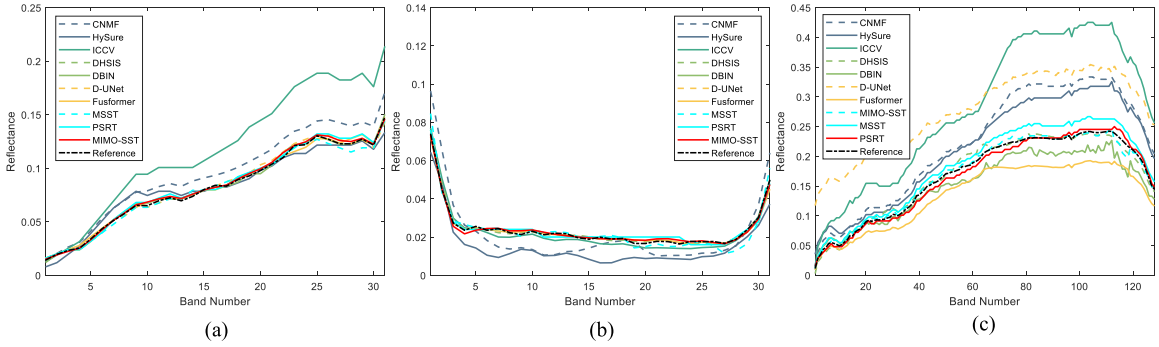


Fig. 14. Spectral curves in the reconstructed HR-HSIs. (a) Oil painting. (b) rsh_0406-1356. (c) Chikusei dataset.

5) Impact of Super-Resolution on Subsequent Applications: This part can be found in the Supplementary Material.

The proposed method in this article has advantages over other methods in the reconstruction of images at different scales and the reconstruction of global features due to the use of multi-input multi-output methods and the mixture Transformer module. In addition, our method introduces a loss for detail extraction, providing an advantage in detail reconstruction compared to other methods in the reconstructed images. Although the CTB module that we proposed is a channel-based Transformer, the improvement in spectral reconstruction capability for images is not significant based on the SAM metric. The method proposed in this article performs well in subsequent applications, particularly in classification tasks. This indicates the effectiveness of the method in preserving crucial image features to support effective classification tasks. Therefore, we believe that the images generated by this super-resolution method are not only visually clearer but also provide accurate and distinctive information for classification tasks. This is a valuable characteristic, especially in various

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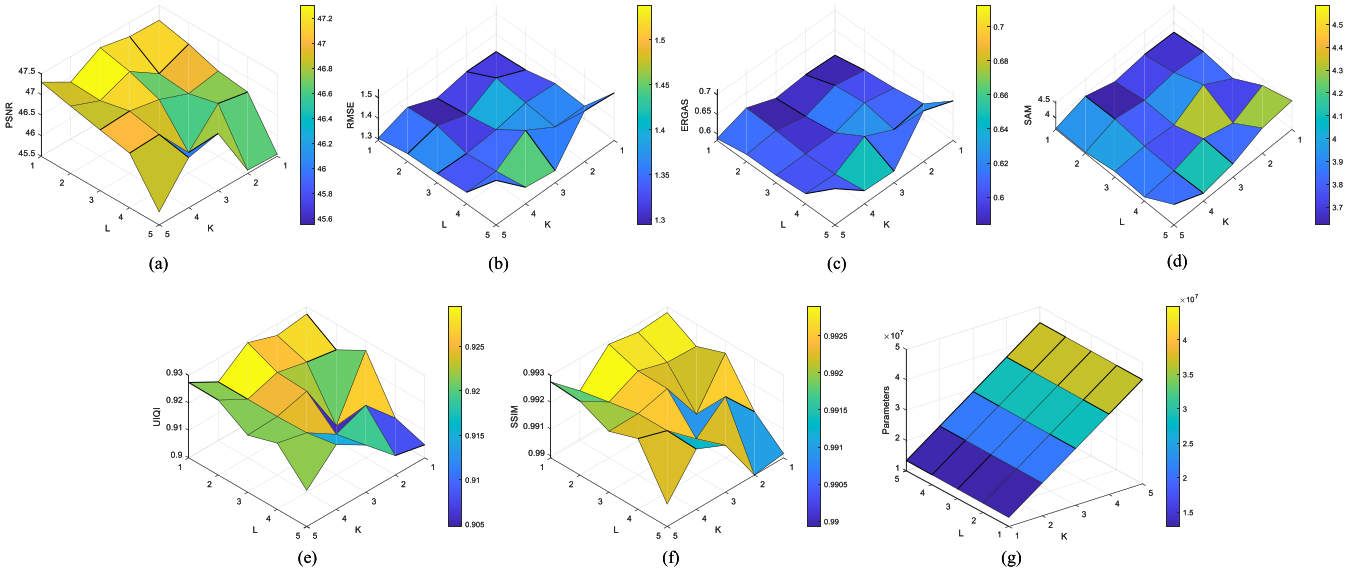


Fig. 15. This study conducted a parametric analysis of the CAVE dataset, evaluating various performance metrics. The metrics analyzed included (a) PSNR, (b) RMSE, (c) ERGAS, (d) SAM, (e) UIQI, (f) SSIM, and (g) parameters.

TABLE VI

THIS STUDY AIMED TO EVALUATE THE PERFORMANCE OF EACH INDIVIDUAL MODULE PROPOSED WITHIN OUR MODEL THROUGH EXPERIMENTATION ON THE CAVE DATASET

MTB	CTB	MIMO	WHF	PSNR	RMSE	ERGAS	SAM	UIQI	MSSIM
				46.0883	1.5247	0.6983	3.9632	0.9265	0.9906
✓				46.7268	1.4035	0.6365	3.8162	0.9263	0.9916
	✓			46.2496	1.4537	0.6556	4.4582	0.9080	0.9903
		✓		45.9596	1.5362	0.7035	4.1462	0.9218	0.9903
✓	✓			47.1341	1.3363	0.5977	3.7704	0.9280	0.9924
	✓	✓		46.4050	1.4116	0.6479	4.9120	0.8961	0.9885
✓		✓		46.4881	1.4234	0.6475	3.9214	0.9246	0.9914
✓	✓	✓		47.1503	1.3183	0.5962	**3.6200**	0.9281	0.9928
✓	✓	✓	✓	**47.3029**	**1.2942**	**0.5838**	3.6227	**0.9295**	**0.9929**

TABLE VII

THIS STUDY SOUGHT TO INVESTIGATE THE IMPACT OF VARYING INPUT SIZES, AS REPRESENTED BY NETWORK WIDTH C, ON NETWORK PERFORMANCE OF THE CAVE DATASET

Dim	PSNR	RMSE	ERGAS	SAM	UIQI	SSIM
C = 48	**47.3029**	**1.2942**	**0.5838**	**3.6227**	**0.9295**	**0.9929**
C = 64	47.1954	1.3096	0.5884	3.7986	0.9232	0.9923
C = 80	46.7028	1.3703	0.6098	3.6727	0.9261	0.9927

TABLE VIII

THIS STUDY AIMED TO COMPARE THE PERFORMANCE OF OUR PROPOSED MCGFN MODULE WITH THAT OF A CONVENTIONAL FN

Component	PSNR	RMSE	ERGAS	SAM	UIQI	MSSIM
FN [50]	46.4742	1.4389	0.6480	4.6058	0.9128	0.9903
GDFN [58]	47.1053	1.3281	0.6004	3.9061	0.9242	0.9923
MCGFN	**47.3029**	**1.2942**	**0.5838**	**3.6227**	**0.9295**	**0.9929**

application scenarios within the fields of image processing and computer vision.

TABLE IX

PERFORMANCE OF THE PROPOSED METHOD WITH VARIOUS NUMBERS OF INPUTS AND OUTPUTS

Number	PSNR	RMSE	ERGAS	SAM	UIQI	MSSIM
1	46.9837	1.3531	0.6087	3.7630	0.9258	0.9923
2	46.9041	1.3692	0.6133	3.9873	0.9213	0.9917
3	**47.3029**	**1.2942**	**0.5838**	**3.6227**	**0.9295**	**0.9929**

TABLE X

ANALYZING THE EFFECTS OF DIFFERENT HIGH-FREQUENCY INFORMATION EXTRACTION OPERATORS

Component	PSNR	RMSE	ERGAS	SAM	UIQI	MSSIM
discrete cosine	47.1576	1.3214	0.5939	3.7522	0.9259	0.9925
Fourier	47.1353	1.3167	0.5932	3.8008	0.9246	0.9923
wavelet	**47.3029**	**1.2942**	**0.5838**	**3.6227**	**0.9295**	**0.9929**

D. Ablation Study

In order to evaluate the efficacy of our proposed MIMO-SST model, we conducted an extensive ablation study on the CAVE dataset. This ablation study involved a comprehensive analysis of several distinct components of the model to gain a deeper understanding of its underlying performance characteristics.

1) Effectiveness of the Proposed Module: An extensive evaluation of the MIMO-SST model was conducted on the CAVE dataset in order to analyze the effectiveness of each individual component. The proposed model is tested with and without MTB, CTB, MIMO, and WHF, with a baseline model trained without these components being used for comparison purposes. The results provided in Table VI indicate that the integration of MTB led to a PSNR improvement of 0.64 dB compared to the baseline model, whereas the impact of CTB alone is found to be relatively modest, resulting in a PSNR increase of only 0.16 dB. However, when combined with

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TABLE XI
RESULTS OF MODELS TRAINED ON THE CAVE DATASET AND TESTED ON THE ICVL DATASET

Dataset	Indices	CNMF [17]	HySure [19]	CSU [23]	DHSIS [26]	DBIN [30]	D-UNet[46]	Fusformer [34]	MSST [35]	PSRT [36]	MIMO-SST
ICVL	MPSNR↑	—	—	—	46.8428	43.7418	47.5401	43.5724	44.1246	47.4258	**48.3946**
	RMSE↓	—	—	—	1.2708	2.2837	1.1895	2.2456	1.9531	1.1961	**1.0497**
	ERGAS↓	—	—	—	0.9017	1.5132	0.9124	1.8687	1.2621	0.9325	**0.7675**
	SAM↓	—	—	—	1.7581	2.6010	1.7433	**1.4621**	2.3727	1.6891	1.5613
	UIQI↑	—	—	—	0.9318	0.9235	0.9268	0.9423	0.9226	0.9366	**0.9393**
	MSSIM↑	—	—	—	0.9904	0.9893	0.9902	0.9923	0.9907	0.9921	**0.9934**

TABLE XII
RUNTIMES, FLOPS, AND NETWORK PARAMETERS OF VARIOUS ALGORITHMS ON THE CAVE DATASET

Indices	CNMF [17]	HySure [19]	CSU [23]	DHSIS [26]	DBIN [30]	D-UNet [46]	Fusformer [34]	MSST [35]	PSRT [36]	MIMO-SST
Times(s)	19.5312	389.7733	164.9055	6.1281	36.0000	1.0200	6.8043	5.1295	4.6332	**0.6020**
FLOPs(G)	—	—	—	1599	8019	421	921	494	630	**420**
Parameters(M)	—	—	—	8.3125	10.2093	16.0099	5.1964	6.143	5.7355	**4.9830**

TABLE XIII
SELECTION OF THE PARAMETER λ

λ	PSNR	RMSE	ERGAS	SAM	UIQI	MSSIM
0.1	47.0237	1.3385	0.6056	4.0228	0.9196	0.9920
0.01	**47.3029**	**1.2942**	**0.5838**	**3.6227**	**0.9295**	**0.9929**
0.001	47.0317	1.3453	0.6089	4.0696	0.9159	0.9916



(a) (b)

Fig. 16. Pseudocolor images of (a) MSIs (bands: 3, 2, and 1) and (b) HSIs (bands: 20, 5, and 3).

MTB, CTB contributed an additional 0.41 dB improvement in performance. Though the inclusion of MIMO led to a decrease in PSNR by 0.13 dB relative to the baseline model, its integration with CTB resulted in a net improvement of 0.32 dB. Ultimately, the use of all three components (MTB, CTB, and MIMO) resulted in a PSNR improvement of 1.07 dB. Finally, the incorporation of WHF yielded a significant PSNR increase of 1.22 dB compared to the baseline model.

2) Wider Network: The performance of the proposed method was evaluated across a range of network widths, as depicted in Table VII. The results suggest that the method does not perform optimally for wider networks (i.e., larger C). The table indicates that the proposed method achieved the best performance in terms of PSNR, RMSE, ERGAS, SAM, UIQI, and MSSIM metrics at $C = 48$. As a result, this value is chosen as the optimal width for the network architecture employed in the proposed method.

3) Improvements in FN: Table VIII presents a comparative analysis of the proposed MCGFN against conventional FN [50] and GDFN [58]. The results indicate that the incorporation of the gating mechanism and different convolution sizes in the conventional FN led to an improvement of 0.83 dB compared to the baseline model. Meanwhile, the proposed MCGFN yielded a comparatively modest increase of 0.20 dB compared to GDFN. Notably, the best performance outcomes were observed in metrics such as RMSE, ERGAS, and MSSIM. In addition, the MCGFN demonstrated superior performance relative to GDFN in SAM and UIQI metrics. In summary, it is shown that the MCGFN proposed in this article is effective.

4) Selection of Hyperparameters K and L: In this section, we investigate the effect of the parameters K and L and their influence on the quality indicators of the proposed method. In this experiment, K is chosen from {1, 2, 3, 4, 5} and L from {1, 2, 3, 4, 5}. Fig. 15(a)–(g) depicts the PSNR, RMSE, ERGAS, SAM, UIQI, SSIM, and parameters of the reconstructed CAVE dataset as a function of K and L, respectively. It can be seen from Fig. 15(g) that an increase in K is a greater increase for the number of parameters in the model compared to an increase in L. Furthermore, the graph shows that RMSE, ERGAS, and SAM are better when $K = 3, 4$ and $L = 1, 2$. Therefore, in combination with the number of parameters and the method proposed in this article, K and L are eventually set to 3 and 1.

5) Performance of the Proposed Method With Various Numbers of Inputs and Outputs: In this section, we compared the effects of using different numbers of inputs and outputs on the experimental results of the CAVE dataset. The quantity being equal to 1 indicates that one LR-HSI and one HR-MSI are input, and one HR-HSI is output. The experimental results are shown in Table IX. It can be observed from the table that our proposed method outperforms other methods in all evaluation metrics. Due to limitations in hardware and other environmental factors, it was not possible to conduct experiments with quantities exceeding three. Therefore, we believe that setting the quantity to three is the optimal choice.

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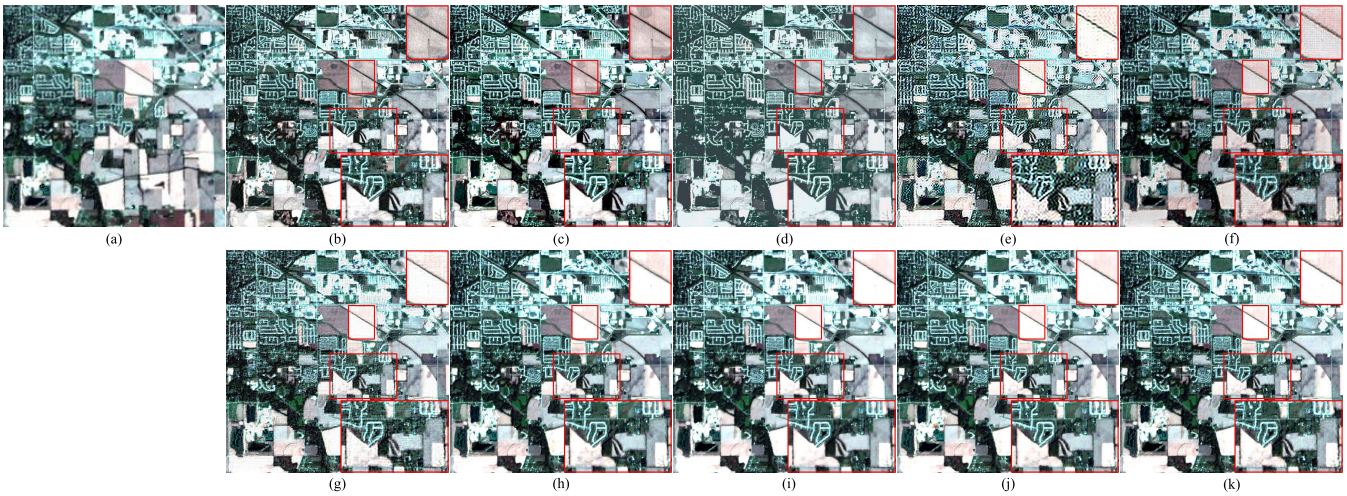


Fig. 17. Fused pseudocolor composite image of Hyperion-sentinel results (bands: 20, 5, and 3); the size of the image is 600 × 600 with a 10-m resolution. (a) LR-HSI. (b) CNMF [17]. (c) HySure [19]. (d) CSU [23]. (e) DHSIS [26]. (f) DBIN [30]. (g) D-UNet [46]. (h) Fusformer [34]. (i) MSST [35]. (j) PSRT [36]. (k) MIMO-SST.

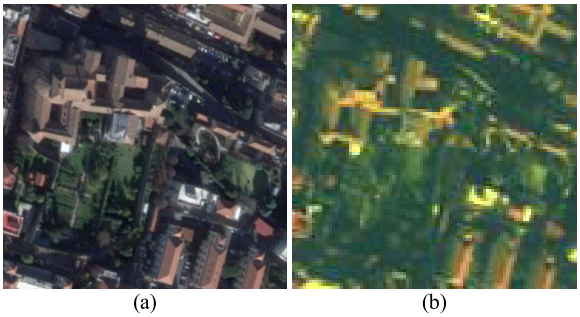


Fig. 18. Pseudocolor images of (a) RGBs and (b) MSIs (bands: 5, 3, and 1).

6) Examining the Impact of Various High-Frequency Information Extraction Operators: In this section, we compared the effects of using the discrete cosine transform (DCT) and Fourier transform methods for extracting high-frequency information on the CAVE dataset with our proposed wavelet-based method. The experimental results are presented in Table X. It can be observed from the table that our proposed method achieves the best results among other methods in all evaluation metrics. Therefore, we can conclude that incorporating the wavelet method in the network for high-frequency information extraction is effective.

7) Generalization of the Proposed Method Is Validated: In this section, we conducted tests on the ICVL dataset using a model trained on the CAVE dataset to verify the generalization performance of our proposed method. The experimental results are shown in Table XI. From the table, it can be observed that our proposed method achieves the best results in terms of evaluation metrics, such as PSNR, RMSE, ERGAS, UIQI, and SSIM, compared to other methods. It falls slightly behind the Fusformer method only in the SAM metric. Thus, we can conclude that our proposed method exhibits favorable generalization performance.

8) Runtime Analysis, Floating-Point Operations, and Network Parameter Analysis: Table XII presents the quantitative comparison of the average running times and floating-point operations per second (FLOPs) of different methods for 8× super-resolution processing on the CAVE dataset. By observing the table, the proposed method achieves excellent results regarding running time, FLOPs, and network parameters. Despite incorporating a global spatial attention mechanism, our model has a smaller number of modules. This will result in a lightweight model, thereby reducing testing time.

9) Selection of Parameter λ: In this section, we investigated the impact of the parameter on the quality indicators of the proposed method. In this experiment, we selected λ from the set {0.1, 0.01, 0.001}. It can be observed from Table XIII that the performance of the proposed method is optimal when $\lambda = 0.01$. Therefore, we ultimately set λ to 0.01.

V. Real Data Experiment

In the following, the real dataset is used to verify the effectiveness of the proposed methods further. We use the LR-HSI acquired by Hyperion sensors using the Eo-1 satellite and the HR-MSI acquired using the Sentinel-2 satellite. The Hyperion HSI has a spectral range of 400–2500 nm, including 242 bands, with a spatial resolution of 30 m. After removing the water vapor and noise bands from the HSI, 89 bands remain. The MSI S2 has a total of 13 bands, from which four bands of 490, 560, 665, and 842 nm are selected as the HR-MSI with a spatial resolution of 10 m.

This section aims to fuse 30-m HSI and 10-m MSI data to obtain 10-m HSI. Since the proposed network is supervised learning, 10-m HSI data are required as a reference image but no 10-m HSI data in the real scene. Therefore, we use the strategy to convert 30-m HSI and 10-m MSI through downsampling. The original 30-m HSI is used as a reference image for training, while, in the testing phase, the original 30-m HSI and 10-m MSI are used to obtain 10-m HSI data.

The Hyperion HSI contains a spatial size of 2350×990, and the size of the S2 MSI is 7050×2970. For the experiment, 200×200 pixels from the HSI data and 600×600 pixels from the MSI data are cropped as the test set. At the same time, the rest of these data are used as the training dataset. Furthermore,

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TABLE XIV
BRISQUE COMPARISONS ON THE REAL DATASET

Indices	dataset	CNMF [17]	HySure [19]	CSU [23]	DHSIS [26]	DBIN [30]	D-UNet [46]	Fusformer [34]	MSST [35]	PSRT [36]	MIMO-SST
BRISQUE↓	Hyperion-sentinel	7.2534	25.0412	19.3943	42.7885	5.0287	17.1247	42.2798	17.4309	30.8849	15.7573
	Worldview-2	35.6627	37.0717	37.6130	39.6349	35.0606	33.2624	37.1994	34.6130	37.4322	**26.2765**

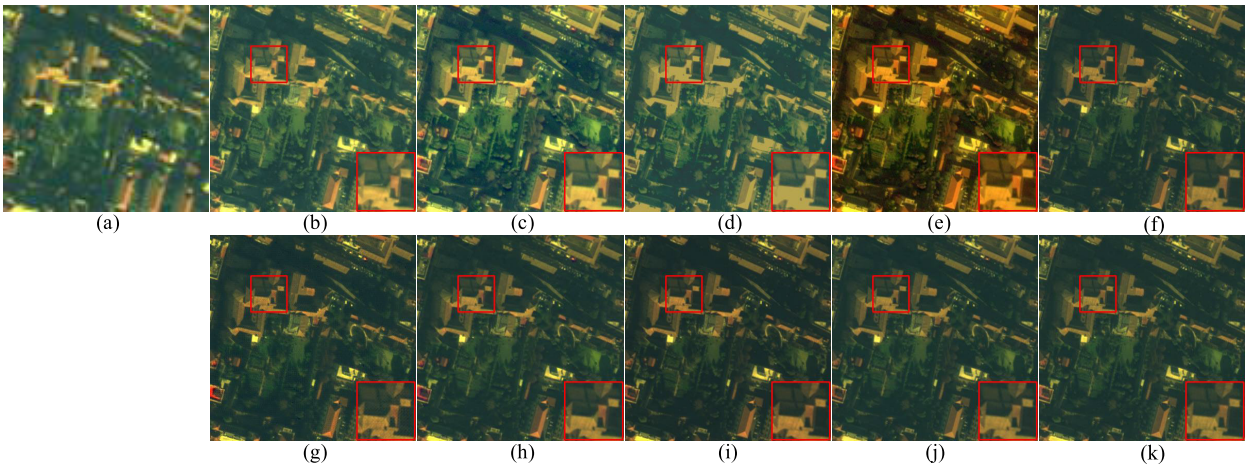


Fig. 19. Fusion result of Worldview-2's pseudocolor composite map (bands: 5, 3, and 1); the size of the image is 400 × 400. (a) LR-HSI. (b) CNMF [17]. (c) HySure [19]. (d) CSU [23]. (e) DHSIS [26]. (f) DBIN [30]. (g) D-UNet [46]. (h) Fusformer [34]. (i) MSST [35]. (j) PSRT [36]. (k) MIMO-SST.

the test images are shown in Fig. 16. The training image is divided into patches with the $4 \times 4 \times 89$, $12 \times 12 \times 4$, and $12 \times 12 \times 89$ for the LR-HSI, HR-MSI, and HR-HSI, respectively. Since the Hyperion-Sentinel dataset is unlabeled, quantitative assessment of the generated image quality can only be done using nonreference metrics. Because the Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE) [71] method is effective in assessing the spatial quality restoration outcomes of nonreference natural images, we have chosen to employ this method to evaluate the spatial quality of the generated HSI.

From Fig. 17, it can be seen that in the fused images generated by the CNMF, Hysure, and CSU methods, the spectra of these images are more multispectral, not taking into account that the spectra of the fused images should be more in the HSIs and that the CSU method has a kind of spectral distortion in the whole fused image. The DHSIS method has streaks of noise in the fused image, and the fused image is blurred, while the DBIN method has more texture noise in the fused image. The D-UNet method has blurred edges and noise in the fused image. The images fused using the Fusformer, MSST, and PSRT methods exhibited excessive smoothness, leading to a notable loss of image details. Table XIV presents a quantitative comparison using the BRISQUE metric on the Hyperion-Sentinel dataset. From the table, it can be observed that the values for CNMF and DBIN methods are smaller than those for our proposed method, indicating that these two methods perform better in terms of spatial reconstruction. However, as shown in Fig. 17, CNMF and DBIN methods exhibit larger spectral reconstruction errors compared to LR-HSI in the top-right corner of the image reconstruction. Therefore, taking into account both Fig. 17 and Table XIV, we conclude that our proposed method outperforms other methods in both spatial and spectral reconstructions on the Hyperion-Sentinel dataset.

Moreover, we conduct performance validation of the proposed method on the real-world MSI dataset WV2.[12] This dataset comprises pairs of actual LR-MSI with eight bands and HR-RGB images. We perform HR-MSI reconstruction from LR-MSI and HR-RGB pairs. As the dataset contains only pairs of images, we initially designate the 100×100 RGB and 400×400 MSI data as the test set, with the remaining data serving as the training dataset. The test set data are illustrated in Fig. 18. Similarly, we employ the block method to prepare the training dataset, where the sizes of LR-MSI, HR-RGB, and HR-MSI image blocks are $3 \times 3 \times 8$, $12 \times 12 \times 3$, and $12 \times 12 \times 8$, respectively.

As shown in Fig. 19, the fused images generated by CNMF and HySure methods exhibit blurriness and excessive brightness. The CSU method results in spectral distortion, particularly noticeable in red roofs against a gray background. The DHSIS method generates images with color distortion, while images generated by the DBIN and D-UNet methods appear blurry. The roof colors in images generated by Fusformer and MSST methods are distorted. The image generated by the PSRT method is closest to our proposed method. Table XIV provides a quantitative comparison using the BRISQUE metric on the Worldview-2 dataset. From the table, it can be observed that our proposed method performs the best in terms of spatial reconstruction. Therefore, we conclude that our proposed

[12]https://www.harrisgeospatial.com/Data-Imagery/Satellite-Imagery/High-Resolution/WorldView-2

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method outperforms other methods in both spatial and spectral reconstructions on the Worldview-2 dataset.

Based on the above analysis, it can be concluded that MIMO-SST exhibits superior spatial and spectral reconstruction capabilities on real datasets.

VI. Conclusion

Current sensors are unable to simultaneously acquire images with high spatial resolution and high spectral resolution. As a result, existing methods rely on the fusion of LR-HSI and HR-MSI. While existing end-to-end Transformer-based techniques can extract global information from HSI, they may not perform optimally for tasks that emphasize local information, such as HSI fusion. Therefore, this article proposes a novel network architecture called MIMO-SST. The use of multi-input multi-output methods facilitates supervised HSI generation across various scales, allowing for a more comprehensive capture of details and structures in the images. To apply multi-input multi-output methods in HSI fusion, specifically integrating the spatial-spectral information of LR-HSI and HR-MSI, we employ MFMA, MFCA, and a multiscale convolutional gated FN to construct the proposed hybrid SST. Furthermore, to enhance the network's focus on edge information in the images, a novel WHF loss is integrated into the final total loss. This integration aims to improve the reconstruction of high-frequency details.

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Jian Fang received the B.S. degree in computer science and technology from Hefei University, Hefei, China, in 2017. He is currently pursuing the Ph.D. degree in computer science with the Nanjing University of Science and Technology (NJUST), Nanjing, China.

His fields of interest include deep learning and remote sensing image processing.

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Jingxiang Yang (Member, IEEE) received the Ph.D. degree in control theory and control engineering from Northwestern Polytechnical University, Xi'an, China, in 2019, and the Ph.D. degree in engineering science from Vrije Universiteit Brussel, Brussels, Belgium, in 2019.

He is currently a Lecturer with the Nanjing University of Science and Technology, Nanjing, China. His research interests include deep learning and its applications in hyperspectral image processing.



Abdolraheem Khader (Member, IEEE) received the B.S. degree in computer science from Karary University, Omdurman, Sudan, in 2011, the M.Sc. degree in computer science from the Sudan University of Science and Technology, Khartoum, Sudan, in 2014, and the Ph.D. degree in computer science from the Nanjing University of Science and Technology, Nanjing, China, in 2023.

He is currently a Post-Doctoral Researcher with the Nanjing University of Science and Technology. His research interests include the areas of deep learning and hyperspectral image super-resolution.



Liang Xiao (Senior Member, IEEE) received the B.S. degree in applied mathematics and the Ph.D. degree in computer science from the Nanjing University of Science and Technology (NJUST), Nanjing, China, in 1999 and 2004, respectively.

From 2009 to 2010, he was a Post-Doctoral Fellow with the Rensselaer Polytechnic Institute, Troy, NY, USA. Since 2014, he has been the Deputy Director of the Jiangsu Key Laboratory of Spectral Imaging Intelligent Perception, Nanjing. He was the Second Director of the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information of the Ministry of Education, NJUST, where he is currently a Professor with the School of Computer Science. He has published more than 100 international journal articles, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. His main research interests include inverse problems in image processing, computer vision and image understanding, pattern recognition, and remote sensing.

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